

Named Entity Extraction for Food Safety Events Monitoring



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Authorship Attribution Statement

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- I prepared the manuscript.
- Dr. R. Chen provided the content for the Online Food Safety News Monitoring component.
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- I provided the content for the Named Entity Extraction for Food Safety Monitoring component, as reported in this thesis.
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Summary

Due to the dynamic nature of the food supply ecosystem and the rising reports on food-borne disease outbreaks, effective food safety risk monitoring has become crucial for public health and socioeconomic development. The Singapore Food Agency (SFA) is a statutory board that oversees the food safety and security of Singapore. Part of its functions is to monitor and detect potential food-borne incidents. Currently, SFA scientists rely on manual efforts to gather relevant information from various internal datasets to conduct food safety analysis, risk assessments, and decision making. This is not only time-consuming but also susceptible to human errors. To tackle this problem, the Centre for Smart Platform Infrastructure Research on Integrative Technology (SPIRIT), NTU and SFA have collaborated to develop a predictive food safety risk monitoring system called SFA-FSM. The SFA-FSM system applies advanced analytics and artificial intelligence (AI) technologies for monitoring online food safety news and local customer feedback.

This thesis proposes a Named Entity Extraction for Food Safety Monitoring (NEE-FSM) framework as a crucial component for information extraction (IE) to support multiple downstream functionalities of the SFA-FSM system. The NEE-FSM framework introduces a hybrid approach for dictionary-based named entity extraction, consisting of three modules: the Heuristic Entity Extraction (HEE) module, which constructs base dictionaries of commonly known food safety named entities from various knowledge resources; the Named Entity Recognition (NER) module, which iteratively augments and enhances the base dictionaries with new named entities from the food safety datasets; and the Named Entity Matching

(NEM) module, which performs dictionary-based named entity extraction using the augmented dictionaries.

This thesis focuses on discussing the various techniques to incorporate publicly available knowledge resources to effectively model the components of the NEE-FSM framework in low-resource settings. The proposed framework has shown to be effective for food safety named entity extraction, as demonstrated by the performance results of our experiments.

Chapter 1

Introduction

1.1 Background

The consumption of unsafe food carrying food-borne hazards such as harmful bacteria, viruses, parasites, and chemical substances, can have significant impact on the health of individuals, increasing the risk of developing illnesses and diseases from immediate effects such as diarrhoea to long-term effects such as cancer. At a larger scale, outbreaks of food-borne diseases can contribute to national or global-level food safety issues, impeding socioeconomic development by exhausting medical and healthcare systems, and harm economies, tourism and trades [1].

Statistics have shown a rise in global reports of approximately 1.5 billion food-borne disease outbreak cases, that have resulted in 3 million deaths annually [2]. According to the World Health Organization (WHO), due to the dynamic nature of the food supply ecosystem and its impact on food safety, the importance for food safety monitoring has become increasingly crucial to protect the well-being of public health from potential food-borne hazards, and minimize the risks of food-borne disease outbreaks [1]. The Singapore Food Agency

(SFA), in collaboration with the food industry to grow food supply, seeks to maintain Singapore as one of the lowest counts for food safety incidents globally, despite 90% of the consumed food are imported. Through stringent standards and operations, SFA oversees food safety and detects potential food-borne incidents through physical inspections of food products and establishments, investigating customers' feedback, conducting laboratory tests on food samples, and the monitoring of global food safety alerts [2].

With the increasing complexity in food safety monitoring and risk assessments, SFA faces the following challenges:

- SFA scientists currently rely on manual efforts to gather relevant information from various internal datasets for food safety risk assessments and decision making process to inform follow-up actions on the implicated food products and establishments. These processes are not only time-consuming which may result in delayed response time, but also susceptible to human errors.
- The information presented across the food supply chain are fragmented, and are often in unstructured texts. Hence, it is difficult to consolidate useful information for holistic analytics on the data.
- External knowledge resources, such as food safety news articles, are publicly available on the Internet. Such resources can provide useful information for global food safety monitoring. However, these are not directly accessible for integration with the existing systems in place.
- The transition to modern day digitization poses challenges as the legacy data, along with the growing volume of new data in different formats and structures, raises the complexity for data pre-processing, information extraction, and analysis.

To address the above challenges, the Centre for Smart Platform Infrastructure Research

on Integrative Technology (SPIRIT), NTU, has collaborated with SFA to develop a Predictive Food Safety Monitoring system called *SFA-FSM*, which applies advanced analytics and artificial intelligence (AI) technologies for holistic food safety monitoring, discovery of emerging risks and hazards, and performing near real-time analytics on potential food-borne incidents to inform SFA operations. The functionalities of the SFA-FSM system components are designed to support and automate the use cases presented by SFA for food safety monitoring including online food safety news monitoring and customer feedback monitoring¹.

1.2 Food Safety Events Monitoring

The key AI technologies for food safety events monitoring in the proposed SFA-FSM system comprises three main components: online food safety news monitoring, food safety feedback monitoring, and named entity extraction for food safety related entities.

1.2.1 Online Food Safety News Monitoring

With increasing concerns about global food safety events, including the rise of food-borne incident outbreaks, global food recalls, and zoonotic diseases that can impact the food supplies, exploring the use of external knowledge resources for detecting global food safety events is a promising approach. Specifically, we monitor online food safety related news articles to extract the relevant information for comprehensive monitoring and analysis of emerging food safety hazards on a global scale.

Figure 1.1 shows the workflow for online food safety news monitoring in the SFA-FSM system. The process begins with web-scraping to download food safety related news articles from specific news websites. Each extracted article content comprises an event title and

¹The work has been published in [3].

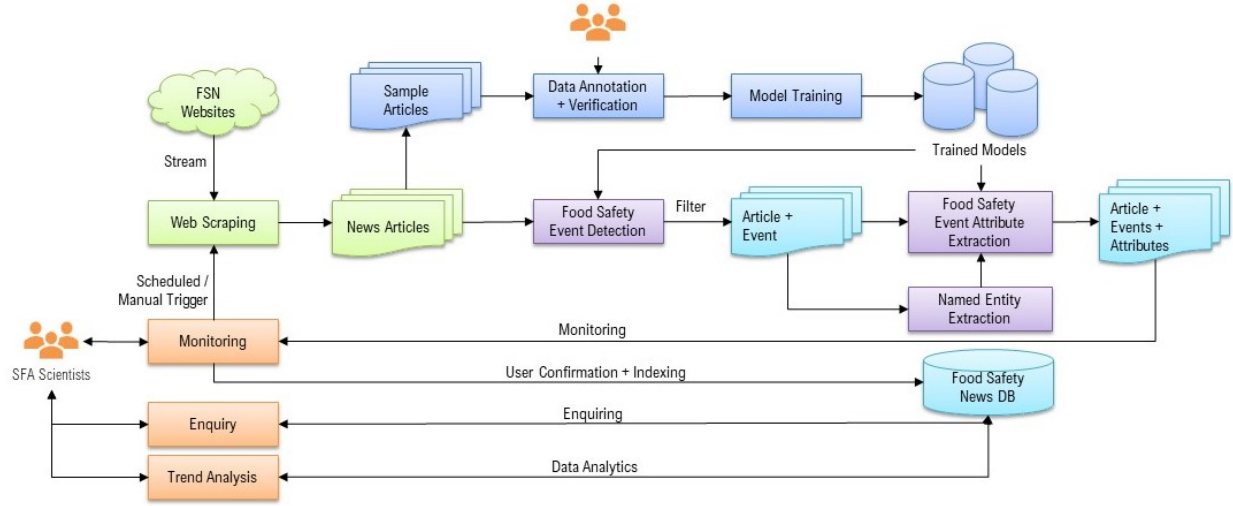


Figure 1.1: Online food safety news events monitoring.

the article body. Additional information, including the article URL and publication date, is downloaded from the web metadata in HTML. For news monitoring, we focus on processing the article contents, which are passed through a food safety event extraction pipeline consisting of two main components, namely Event Detection (ED) and Event Attribute Extraction (EAE). ED is performed on the news articles to identify the presence of three event types: *"Food Poisoning"*, *"Food Recall"*, and *"Zoonotic Disease"*. If no events are detected, the news article is tagged as *"Negative"*. Next, news articles tagged with event types undergo EAE that uses the Named Entity Extraction to identify food safety related entities as attribute candidates, and subsequently classifies these attributes to specific event roles in each food safety event type. For example, attributes such as *"Food"*, *"Causative Agent"*, *"Symptom"*, *"Location"*, and *"Brand"* are extracted for *"Food Poisoning"* events. The news articles, along with the extracted event types and attributes, are processed as new incoming events for global food safety news monitoring in the user interface. Users have the option to modify and confirm these cases to be indexed into the datastore. These information can be accessed through search inquiries by specifying event types, attributes, or specific keywords. Additionally, the information can be processed and analyzed to identify patterns for trend analysis and visualization.

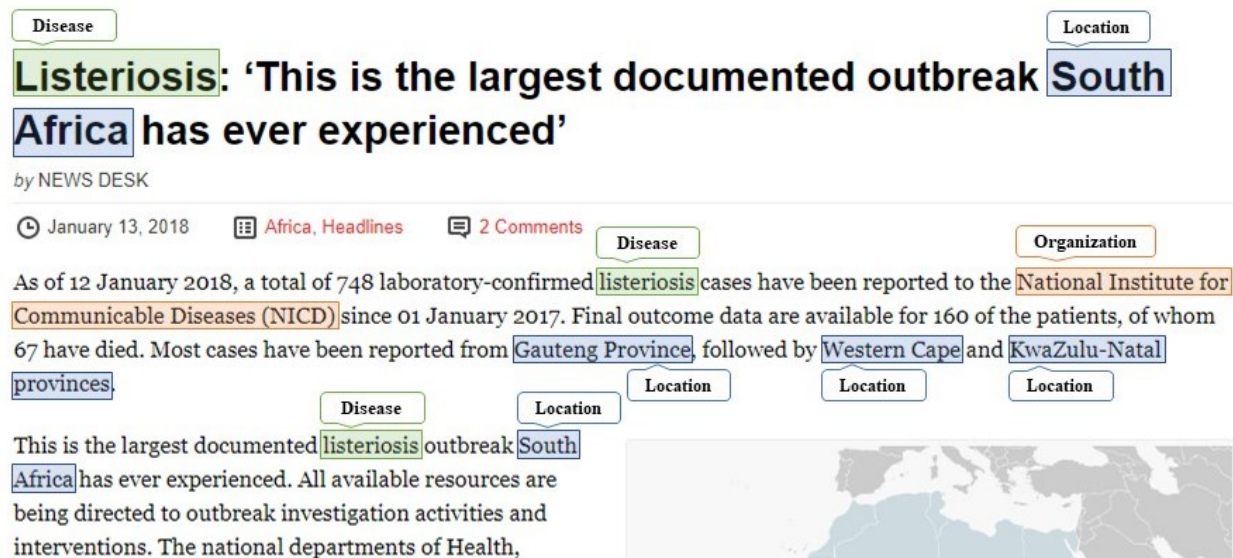


Figure 1.2: Sample food safety news article on food poisoning event.

The online food safety news monitoring mainly involves the event extraction process, which consists of Event Detection (ED) and Event Attribute Extraction (EAE). An example of a food safety news article web-scraped from the Outbreak News Today website is shown in Figure 1.2.

For the ED process, the article title: *"Listeriosis: 'This is the largest documented outbreak South Africa has ever experienced'"* and article contents: *"As of 12 January 2018, a total of 748 laboratory-confirmed listeriosis cases..."*, are merged into a single input sequence. A XLNet pre-trained model [4] is fine-tuned to capture large contextual representations of the entire food safety news article for a multi-labeled classification task. Binary classifiers are used to identify the presence of each food safety event, namely *"Food Poisoning"*, *"Food Recall"*, and *"Zoonotic Disease"*, as an article may involve multiple food safety events. In the example given in Figure 1.2, the news article is classified as a *"Food Poisoning"* event.

For the EAE process, the articles are split into sentence-level for processing. The Named Entity Extraction is employed to extract the named entities and positional information from each sentence as event attribute candidates. For example, given the input sentence

"Listeriosis: 'This is the largest documented outbreak South Africa has ever experienced'", *"Listeriosis"* is extracted as a *"Disease"* named entity, and *"South Africa"* is extracted as a *"Location"* named entity as highlighted in Figure 1.2. Then, a pre-trained BERT [5] model is employed to extract the contextual embedding for each event attribute candidate using the positional information. The extracted embeddings are used in a binary classifier to determine if the candidate plays an event role in the given event. For example, *"Listeriosis"* is classified as the implicated *"Disease"* event attribute, and *"South Africa"* is classified as the implicated *"Location"* event attribute for the *"Food Poisoning"* event. On the other hand, the *"Location"* named entities, namely *"Gauteng Province"*, *"Western Cape"*, and *"KwaZulu-Natal Provinces"*, do not play the implicated *"Location"* role for this specific *"Food Poisoning"* event, based on the context of the sentence *"Most cases have been reported from Gauteng Province, followed by Western Cape and KwaZulu-Natal provinces."*, as these attributes are from previous reported food poisoning cases.

1.2.2 Food Safety Feedback Monitoring

The Customer Relationship Management System (CRMS) dataset contains a collection of user feedback reports. Due to evolving SFA policies and procedures, CRMS feedback reports vary in format. Additionally, CRMS data is collected either through the open-ended online feedback form or manual transcription of phone calls by various SFA officers. Consequently, the feedback texts are largely free-form sentences in natural language without a standard format. With hundreds of new feedback reports received on a weekly basis, it becomes challenging for SFA scientists to manually curate and classify the urgency of these reports, which may lead to delayed operational response. To alleviate manual efforts and improve the scalability for CRMS feedback monitoring, a pre-trained language model is fine-tuned to classify the urgency level of feedback reports, and automate the extraction of important attributes from the feedback texts. The extracted feedback attributes include *"Number of*

people affected", *"Name of symptoms"*, *"Frequency of symptoms"*, *"Anyone hospitalized?"*, and *"Anyone admitted to ICU?"*. The feedback urgency classification of each feedback case will help prioritize the SFA's inspection task in a more efficient and effective manner.

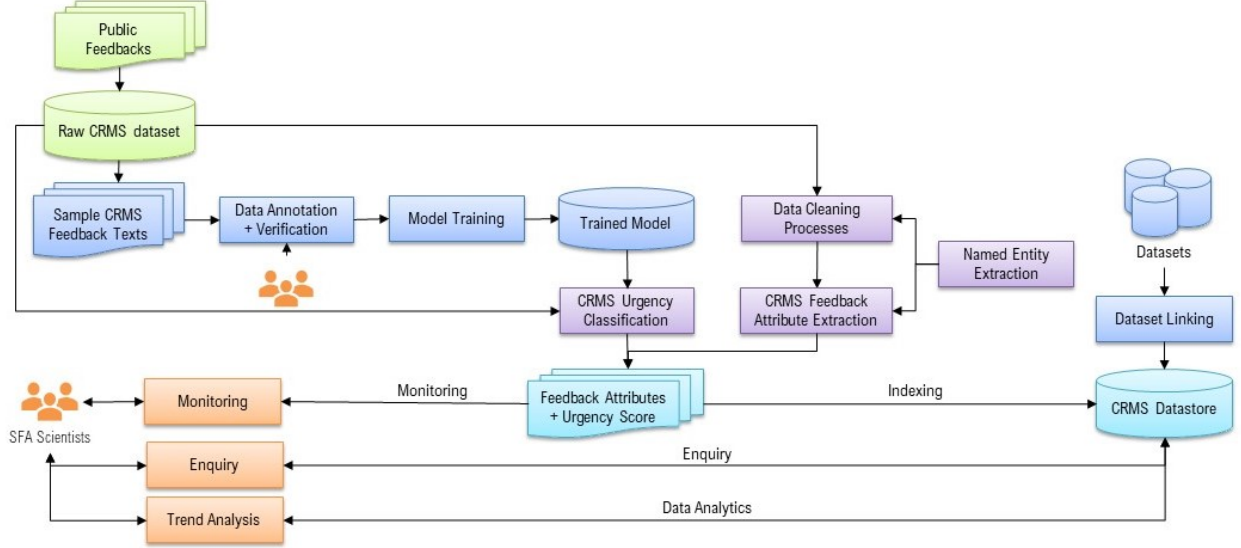


Figure 1.3: Food safety feedback monitoring.

Figure 1.3 shows the workflow of the proposed approach for food safety feedback monitoring for CRMS feedback cases. Each CRMS feedback report contains the feedback category, feedback type, case description, and several other confidential fields. For feedback monitoring, we focus on processing the case description. The process begins with data cleaning of the raw feedback texts using techniques such as spell-checking [6], data filtering, and language translation [7]. The cleaned sentences then undergo the feedback attribute extraction pipeline, where relevant attributes are extracted with the support of Named Entity Extraction. In addition, we also perform feedback urgency classification to predict and assign the urgency or risk level of each feedback case as *"High"*, *"Medium"*, or *"Low"* risk. The CRMS feedback information, along with the extracted attributes and urgency scores, are indexed into the datastore which can be displayed in the user interface. The information can also be accessed through search inquiries with specific attributes or keywords. Further analysis of information can be performed to identify patterns for trend analysis and visualization.

Feedback

Your feedback is of great value to us as it will help us continually improve what we do. Please complete and submit your feedback below.

Feedback Category

Category *

Food Poisoning

Feedback Details

Case Description *

Disease

Location

Food

Caller wants to feedback on a suspect food poisoning at XXX. Caller and friend had ordered the full shell cockles on 19 April from their direct delivery service and requested for the delivery arrival to be 12-1pm. On 20th at about 4-7am, caller start having symptom of bad diarrhea 2 times and vomiting 3 times. Caller had checked with her friend, her friend is ok but the mum had diarrhea.

Symptom

Symptom

You may enter 1 characters and there are 49610 characters remaining.

Figure 1.4: Screenshot of SFA Feedback with sample feedback case description.

The food safety feedback monitoring mainly involves two main components, the urgency classification and the feedback attribute extraction. Figure 1.4 shows a screenshot of the SFA feedback form for CRMS, with an example of the customer feedback report extracted from the CRMS dataset. The details include a "Food Poisoning" feedback category, with the written case description.

For urgency classification, the case description text: "Caller wants to feedback a suspect food poisoning at XXX. Caller and friend...", is used as the input sequence. A BERT pre-trained model [8] is fine-tuned to capture the contextual representation of the case description. A linear classifier is used to predict and assign the urgency risk level of the feedback case description as "High", "Medium", or "Low" risk.

The feedback attribute extraction consists of three processes, namely data cleaning, data augmentation and attribute extraction. The data cleaning process involves spelling correction for food and symptom names from the case description, with the help of Named Entity Extraction, e.g., "*diahreal*" is corrected to "*diarrhea*". A data filtering process is used to remove location, date-time, ID numbers, email address, URLs, and special characters, to avoid ambiguities when extracting values for "*Frequency of symptoms*" and "*Number of people affected*", e.g. "*12-1pm*" in "... arrival to be *12-1pm*" is modified to "... arrival to be $\langle \text{TIMECONTENT} \rangle$ ". Language translation is applied for non-English food names. The Named Entity Extraction is utilized to extract the "*Symptom*" and "*Food*" names for relevant sentence filtering. As shown in Figure 1.4, "*diarrhea*" and "*vomiting*" are extracted as the "*Symptom*" named entities, and "*full shell cockles*" is extracted as the "*Food*" named entity. In addition, other data cleaning processes such as symbol removal, numerical expression standardization, and lemmatization are also performed. Data augmentation is applied using back translation from various languages, to generate frequent textual patterns to enhance the performance of an Apriori Filter [9] to filter relevant sentences, such as "*caller start having symptom of bad diarrhea 2 times and vomiting 3 times.*" that include "*Symptom*" names with "*diarrhea*" and "*vomiting*", and "*Frequency of symptom*" values with "*2 times*" and "*3 times*".

1.3 Named Entity Extraction for SFA-FSM

As discussed in Section 1.2, Named Entity Extraction is a crucial component for information extraction (IE) in the proposed SFA-FSM system, that supports multiple downstream functionalities such as the food safety attribute extraction process for both online news articles and customer feedback texts. The Named Entity Extraction component aims to extract specific entity mentions from various categories that are related to the food safety domain from the given texts, to specifically cater to the needs of the downstream components. In this

Named Entity Category	Description
Disease	The implicated food-borne or zoonotic-related disease names.
Food	The implicated food product names of the food safety event.
Location	The location names which include regions, countries, provinces, states, and places of the implicated food safety event.
Symptom	The clinical symptom names derived from the implicated disease.

Table 1.1: Named entity categories for food safety related entities.

research, we focus on investigating the Named Entity Extraction component for extracting named entities from four food safety related categories of named entities, namely "*Disease*", "*Food*", "*Location*", and "*Symptom*", which are shown in Table 1.1.

Over the years, Named Entity Extraction, also commonly known as the Named Entity Recognition (NER) task, has been a well-studied problem in the natural language processing (NLP) research. The research communities have introduced various applications of techniques, methodologies, and current practices for the domain specific Named Entity Extraction task, ranging from traditional heuristic rule-based approaches to neural approaches that utilize Transformer-based pre-trained language models [10]. Although heuristic approaches produce high precision in performance, the coverage of the extracted entities is often limited as they need to conform to a finite set of manually defined rules. It is often a challenge to exhaustively enumerate all possible rules. This is particularly true when dealing with more complex sentence structures such as free-formed texts. With the emergence of Transformer-based pre-trained language models [5, 10], neural approaches have produced competitive performance on complex textual structures. However, they may introduce "noisy" extractions that incur additional post-processing cost, especially for domain specific contexts. Therefore, various studies have introduced hybrid approaches [11, 12, 13] that leverage both heuristic and neural approaches to achieve wider coverage while maintaining the quality of extractions [12, 14, 13]. As such, the hybrid approaches are commonly adopted for domain specific NER

tasks to extract highly specialized entity terms from specialized texts.

This research investigates the Named Entity Extraction task for the food safety domain, as a crucial function to support the downstream components of the SFA-FSM system for food safety events monitoring. However, the development of the food safety domain related entity extraction task presents several challenges which are summarized as follows:

- The contents of the two target datasets for the Named Entity Extraction task, namely the online food safety news articles, and the customer feedback reports from the CRMS dataset, contain texts that are different in context, tonality, and structures of the sentences. The news articles texts are lengthy and formally written in well-structured paragraphs, while the customer feedback texts are generally shorter, informally written, including acronyms and short-formed texts, as can be observed from Figure 1.2 and Figure 1.4. Therefore, it is a challenging task to generalize a single NER model as an approach to extract specific entity mentions from the contents of both datasets.
- The utilization of the conventional pre-trained language models such as BERT [5] provides a viable approach for entity extraction modeling, but has its limitations. The food safety domain involves the detection of highly specialized and uncommon terms in the English language, such as "*Disease*" names like "*Clostridium Perfringens Infection*" and "*Salmonella illnesses*", or Singapore location names like "*Ang Mo Kio*" and "*SCC Sembawang Shopping Ctr*". As such, the word distribution in the food safety datasets can differ significantly from the general domain texts that are pre-trained on BERT [5], i.e., Wikipedia and BookCorpus [8, 15]. For instance, the names for food-borne diseases and specific symptoms are rarely found in the general domain texts.
- The food safety domain presents low-resource challenges that are commonly faced by most domain specific NER tasks. This is due to the unavailability of training datasets for training a NER model, typically by applying the conventional supervised approaches

that require a large annotated training dataset. To the best of our knowledge, there are no benchmark datasets available for the Named Entity Extraction task in the food safety domain. Furthermore, the process to manually annotate the datasets from scratch for training is laborious, time-consuming, costly, and heavily reliant on the domain knowledge from the food safety domain.

- The Named Entity Extraction task is required to be highly precise. The capability to capture wide coverage of entities in the food safety categories of interest is important to the performance of the SFA-FSM system. For example, the event attribute extraction task relies on the greedy extractions of named entities as attribute candidates for event role classification.

1.4 Objectives

In this research, the primary objective is to investigate an effective approach for Named Entity Extraction for food safety events monitoring. To achieve this, Figure 1.5 shows the proposed Named Entity Extraction for Food Safety Monitoring (NEE-FSM) framework. The proposed framework consists of three main modules with a man-in-the-middle (MitM) curation process.

Firstly, the Heuristic Entity Extraction (HEE) module constructs comprehensive lists of food safety named entities as the base dictionaries with relatively high quality, from various knowledge resources. Then, the Named Entity Recognition (NER) module extracts named entities from the food safety datasets to augment and enhance the coverage of the base dictionaries. With the help of food safety domain experts from SFA, we introduce a man-in-the-middle (MitM) curation process to ensure the quality of named entities extracted by the HEE and NER modules. Finally, the Named Entity Matching (NEM) module extracts food safety related named entities by matching the entities from the augmented dictionaries

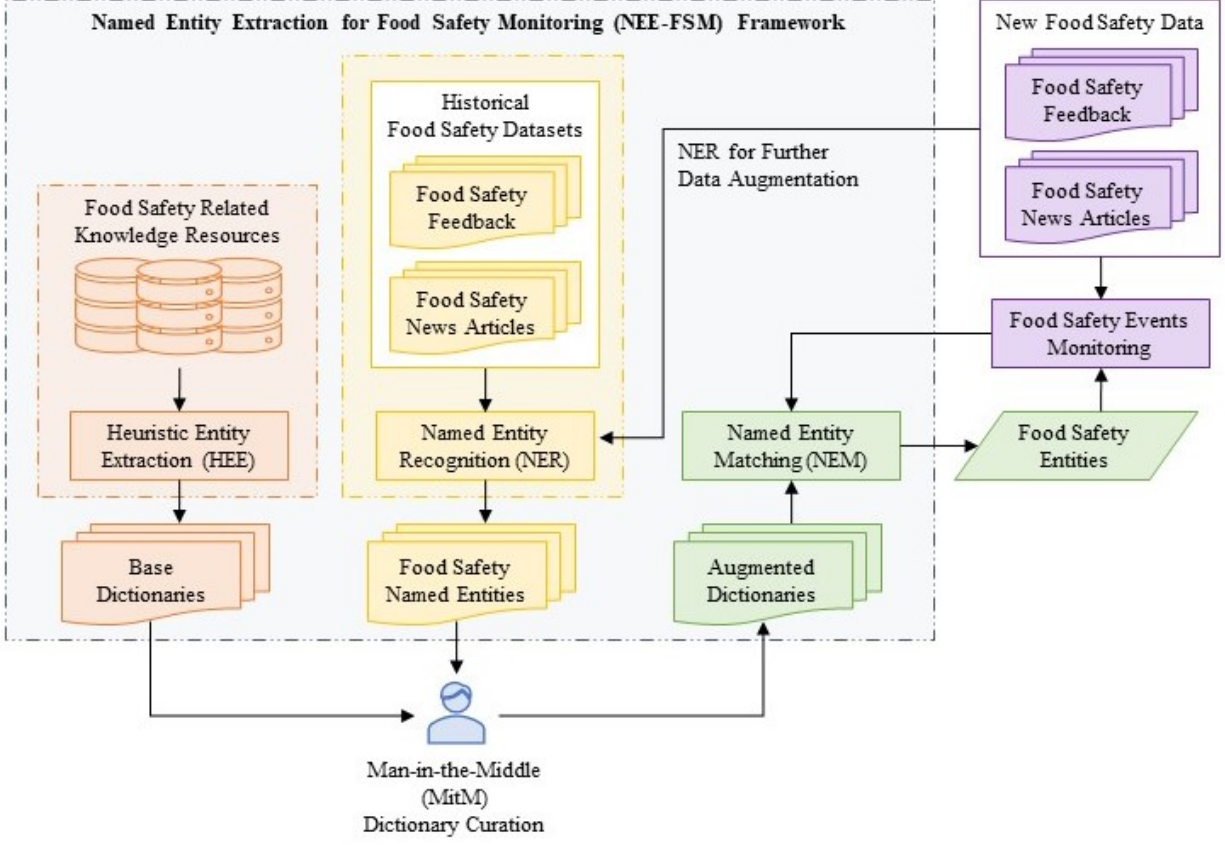


Figure 1.5: Proposed Named Entity Extraction for Food Safety Monitoring (NEE-FSM) Framework.

with the monitored contents of food safety events monitoring. In addition, the NEE-FSM framework allows the Augmented Dictionaries to be continuously updated when new food safety data are processed. Therefore, the iterative updating process can continually improve the overall performance of the NEM process.

More specifically, this research aims to develop the following three modules for the proposed NEE-FSM framework:

- **Heuristic Entity Extraction (HEE)** - It aims to investigate an approach to construct a base set of dictionaries for food safety named entities by leveraging publicly available knowledge resources.
- **Named Entity Recognition (NER)** - It aims to investigate an approach to construct

a neural NER model to extract new "unseen" named entities from the related food safety datasets, to augment the base set of dictionaries to enhance the coverage for entity extraction.

- Named Entity Matching (NEM) - It aims to investigate an approach for dictionary-based named entity matching (NEM) that utilizes the combined lists of named entities from the base dictionaries and NER augmented dictionaries for named entity extraction for food safety events monitoring.

1.5 Contributions

The contributions of this research are summarized as follows:

- Heuristic Entity Extraction (HEE) - This research has proposed various heuristic rule-based approaches, such as Regular Expressions (RegEx) and POS-tagging sequences for pattern matching, for extracting food safety related named entities from the various knowledge resources such as community databases, WordNet, and Wikipedia. As a result, a set of base dictionaries with relatively high quality is created. In total, we have extracted 6,785 commonly known named entities for the specific food safety domain categories, which include 138 "*Disease*" entities, 5,866 "*Food*" entities, 453 "*Location*" entities, and 328 "*Symptom*" entities.
- Named Entity Recognition (NER) - This research has proposed a low-resource named entity recognition (NER) approach to fine-tune pre-trained language models by leveraging multiple NER benchmark datasets from domains that are conceptually related to the food safety domain. It performs transfer learning for named entity extraction on the "*Disease*", "*Food*", "*Location*", and "*Symptom*" categories in the food safety domain. Additionally, we have performed domain adaptive pre-training on the BERT

pre-trained model [5] using unlabeled food safety related corpora to improve the domain adaptive capabilities to capture the contexts of both food safety datasets (i.e., food safety news articles and customer feedback texts). As a result, we can extract more specialized and "unseen" food safety named entities to augment the base dictionaries. In total, we have extracted an additional 1,416 food safety named entities from 200 samples of both datasets. Among them, 948 named entities are extracted from food safety news articles, and 486 named entities are extracted from customer feedback texts, with 18 overlapping named entities. The proposed approaches largely reduce the laborious efforts and cost required to manually annotate the training dataset from scratch.

- Named Entity Matching (NEM) - This research has proposed a lightweight named entity matching (NEM) approach that utilizes heuristic dictionary-based methods to match the food safety related entity mentions with the lists of named entities from the augmented dictionaries. We apply the exact match criterion to match the longest n -gram terms to extract highly precise food safety named entities for the SFA-FSM system. The experimental results have shown promising performance, achieving about 75.15% micro-averaged F1-score on both datasets for named entity extraction. The proposed NEM approach has achieved high precision and significantly improved the performance in recall (about 50.08%) when using the augmented dictionaries compared to the base dictionaries. In addition, the NEM approach has also provided useful support for dataset annotation.

1.6 Organization of This Thesis

The rest of the thesis is organized as follows:

- Chapter 2 reviews the related works for domain specific named entity extraction.

- Chapter 3 discusses the proposed Heuristic Entity Extraction (HEE) module to construct the base dictionaries for the food safety categories of interest.
- Chapter 4 discusses the proposed approaches to construct the Named Entity Recognition (NER) module to extract new "unseen" entities from the target datasets for dictionary augmentation.
- Chapter 5 discusses the proposed approaches for the dictionary-based Named Entity Matching (NEM) module to extract food safety entities.
- Chapter 6 concludes the thesis and discusses possible research directions for future work.

Chapter 2

Literature Review

2.1 Named Entity Recognition (NER)

The named entity extraction, also commonly known as Named Entity Recognition (NER), is a well-studied task that aims to detect and classify named entities from an unstructured sequence of texts into predefined categories of interest, such as "*Location*", "*Organization*", and "*Person*" [16]. NER is widely viewed as the foundational component for information extraction (IE) and text processing, serving multiple downstream applications such as Entity Linking [17], Relation Extraction (RE) [18], Question-Answering (QA) [19], etc.

Over the years, various approaches have been proposed for the named entity extraction task, ranging from heuristic rule-based approaches (e.g., Part-Of-Speech (POS) sequence tagging, dictionary matching, etc.) to statistical machine learning approaches (e.g., Conditional Random Fields (CRF), Hidden Markov Model (HMM), etc.), and deep neural approaches (e.g., Recurrent Neural Networks (RNN), Attention-based Neural Networks, etc.). Moreover, several NER challenges, such as CoNLL'03 [16], ACE'05 [20], TREC-10 Entity Track [21], etc., have also been established by the Natural Language Processing (NLP) research

communities.

2.1.1 Heuristic Approaches

The heuristic rule-based approaches do not require training data as they rely on pre-defined rules or patterns that are handcrafted by domain experts or linguists to identify and extract named entities from a sequence of texts. These rules can be constructed based on lexico-syntactic patterns that exploit linguistic structures [22, 23], such as grammatical patterns from Part-Of-Speech (POS) tagging sequences for extracting noun terms [24], or constructed based on orthographic features from lexicon resources such as gazetteers or dictionaries as look-up tables to exhaustively match distinct names [23]. Sari et al. [25] proposed an approach to exploit grammatical features from Part-Of-Speech (POS) and the syntactical structure of documents to determine the entity boundaries and construct patterns for entity extraction. Pasca et al. [26] proposed lightweight precision-oriented heuristic rules that exploit the Hearst "*Is-A*" patterns [27] for pair-wise extraction of class instances and the associated class from unstructured text.

Although rule-based approaches generally achieve high precision in performance, especially when linguistic patterns are clearly defined in the textual sequence, they lack robustness and commonly suffer from low recall due to the limited set of rules to capture linguistic patterns for more complex variations of textual sequences and incomplete coverage of dictionaries. In addition, it is also challenging and not feasible to enumerate all possible rules [23, 28].

2.1.2 Statistical Machine Learning Approaches

In earlier years, the statistical or feature-based machine learning approaches, that rely on supervised and unsupervised learning algorithms with feature engineering, model named entity extraction as a multi-class classification or a sequence labeling task [22, 28, 23]. The supervised machine learning models, such as Conditional Random Fields (CRF) [29], Hidden Markov Models (HMM) [30], and Support Vector Machines (SVM) [31], learn to extract patterns from input features, such as word-level features (e.g., case, morphology, and Part-Of-Speech (POS) tags) [30], and labeled examples to recognize similar patterns from "unseen" data for extracting named entities [22, 28]. Supervised learning approaches commonly require a large amount of training data due to data sparsity to achieve good performance. The unsupervised machine learning models, typically associated with clustering approaches such as K-Nearest Neighbors (KNN) [32], learn to group the input features into clusters based on contextual similarity by computing the Euclidean distance between features within a vector space [22, 28]. Although unsupervised approaches do not require labeled examples (training data), they rely on well-crafted feature representations of the data to achieve good performance.

Etzioni et al. [33] proposed an unsupervised KNOWITALL system that utilizes named entity candidates as inputs and bootstraps the entity extraction process from a small set of generic extraction patterns. Zhang et al. [34] proposed a token-based tagging system that employs a robust risk minimization classification method and exploits multiple linguistic features, such as word tokens, POS tags, and text chunks, for sequential labeling. Seyler et al. [24] conducted a study on the importance of external knowledge and proposed a framework that incorporates knowledge bases, such as dictionaries and gazetteers, to enhance the input feature representation for training a CRF model for sequential labeling.

2.1.3 Deep Neural Approaches

In recent years, deep neural approaches, such as Recurrent Neural Networks (RNN) [35, 36] and Attention-based Neural Networks (ANN) [10, 8], have emerged as leading approaches in named entity extraction research, achieving competitive state-of-the-art performance. These deep neural architectures consist of multiple processing layers that learn contextual representations from input text sequences. This is typically achieved by forward or backward passing the weighted sum of inputs through a non-linear activation function to discover complex hidden features. Compared to the feature-based machine learning approaches, deep neural approaches do not require intricate feature engineering processes, which are costly and heavily rely on domain experts. Instead, deep neural approaches allow the model to learn contextual representations and underlying features directly from the raw data [22].

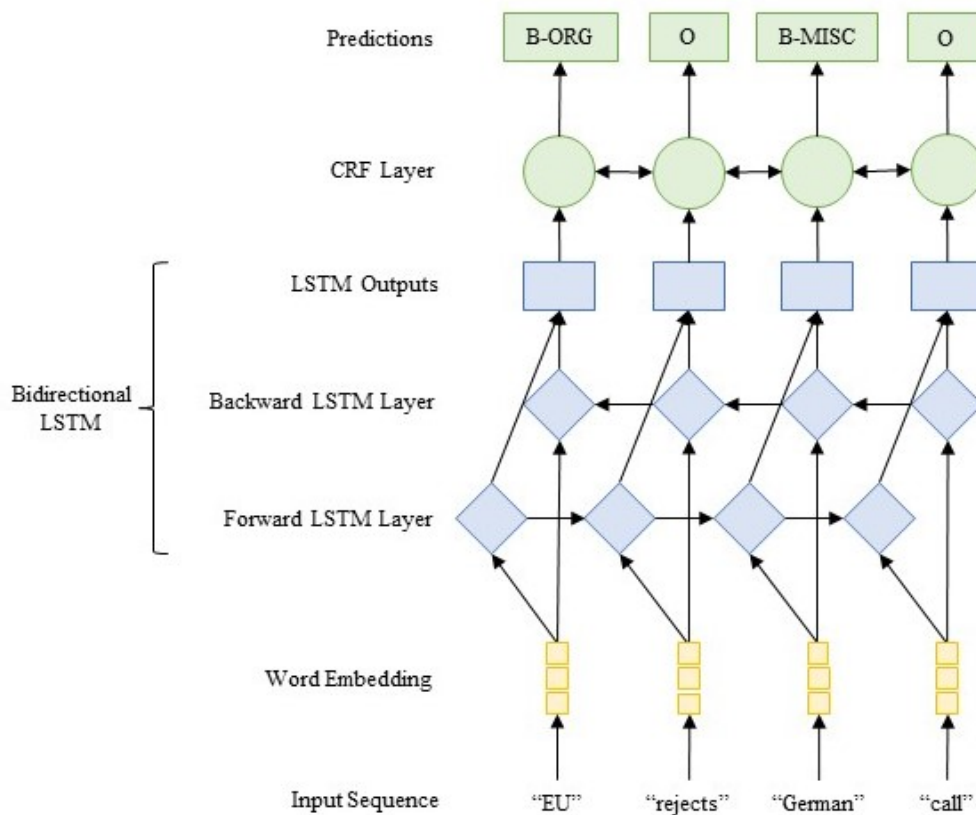


Figure 2.1: Bidirectional LSTM-CRF Model [35].

The biLSTM-CRF model [37, 35, 14, 38, 39] is a well-adopted RNN model used for the named entity extraction task in sequential labeling. As illustrated in Figure 2.1, the model first extracts the word embeddings of the input token sequence. Common processes involve initializing the word embeddings from pre-trained word representation models, such as GloVe [40], Word2Vec [41], and FastText [42]. The word embeddings are then passed into a bidirectional Long Short-Term Memory (BiLSTM) model, which consists of two LSTM layers to capture long-term dependencies in sequential data from both forward and backward directions, improving the contextual understanding of the input sequence. The LSTM layers learn contextual representations by updating hidden states at every step to incorporate information from the past and future context. LSTM utilizes memory cells and gated mechanisms to regulate the flow of information, updating or removing it from the hidden state. The biLSTM outputs concatenate the hidden states of both forward and backward LSTM layers as the contextual embeddings to pass into a Conditional Random Fields (CRF) layer, which considers both contextual information from the biLSTM and relationships among neighboring labels to predict the sequence of labels. Additionally, other works [43, 44, 45] have incorporated Convolution Neural Network (CNN) layers to extract fine-grained contextual information from character-level sequences, capturing meaningful relations and discriminative features between character sequences to enhance the performance of the RNN model and handle terms with complex morphology structures or out-of-vocabulary (OOV) terms.

Although the introduction of LSTM-based RNN models alleviates the potential problem of exploding or vanishing gradients for vanilla RNN models [46], computing more complex textual relationships and contextual dependencies from longer textual sequences may affect the performance of LSTM-based RNN models. The introduction of attention-based neural networks (ANN) [10, 5] proposes a more effective and efficient method for capturing long-range dependencies. They utilize attention mechanisms [10] that compute and assign weights to different tokens in the input sequence based on their importance and relevance. This enables the model to selectively place more "attention" on specific tokens during the

computational phase to capture more complex and fine-grained relations within the input sequence.

2.1.4 Hybrid Approaches

While heuristic rule-based approaches produce highly precise extractions, their coverage is often limited as they adhere to a finite set of pre-defined rules [47]. Conversely, machine learning and deep neural approaches can expand the coverage of named entity extractions but generate noisy results [48, 49, 50]. Therefore, several studies have adopted hybrid approaches that combine machine learning or neural techniques to enhance the extraction coverage while maintaining high precision with rule-based approaches.

Petasis et al. [11] proposed to integrate a feature-based machine learning algorithm with a rule-based NER system to refine the pre-defined rule set. This approach improves performance and reduces manual efforts in maintaining the rule set. The machine learning algorithm is trained on features extracted by rule-based methods (e.g., POS and gazetteer tags). The predictions are compared to rule-based results to identify discrepancies that signal updates to the rule-based system. Srihari et al. [51] combines three modules for named entity and sub-type tagging. It includes a rule-based tagging module that extracts entities such as "time", "date", "percentage", and "monetary expressions" from pattern templates. Next, a gazetteer-based matching module combines gazetteer matches with a Maximum Entropy (MaxEnt) Model to extract "person" and "location" entities. Then, a bi-gram-based Hidden Markov Model (HMM) is utilized for error correction (e.g., entity disambiguation) by leveraging bi-gram statistical information.

2.2 Transformer-based Pre-trained Models

The emergence of the Transformer model, as introduced by Vaswani et al. [10] in their paper "Attention Is All You Need", has gained significant attention in the NLP research community due to its diverse applications in multiple tasks such as Machine Translation, Sentiment Analysis, Text Summarization, Named Entity Recognition, and Question Answering, achieving state-of-the-art performance. The design of Transformer models is entirely based on attention mechanisms, allowing them to effectively capture long range dependencies within textual sequences without relying on recurrent layers.

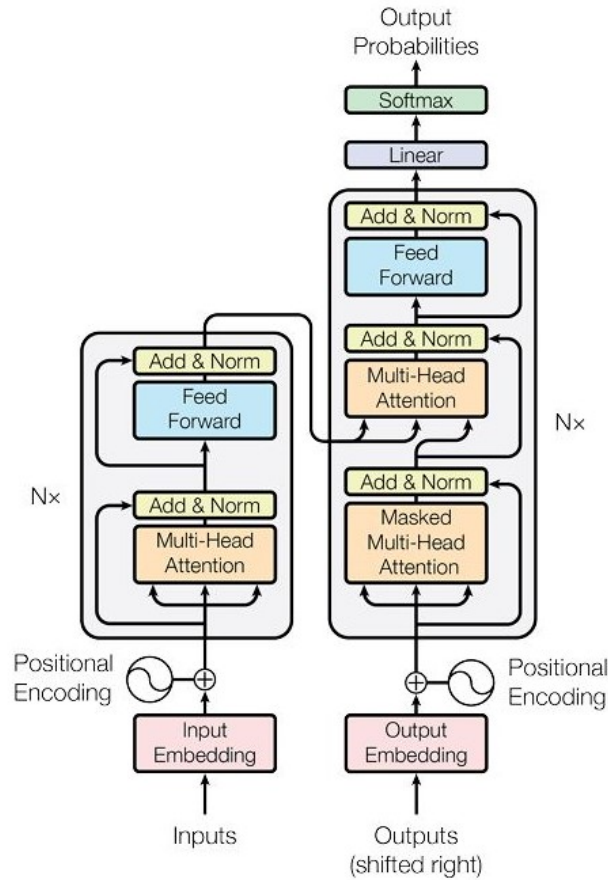


Figure 2.2: Transformer Model Architecture [10].

As shown in Figure 2.2, the Transformer model introduces an encoder-decoder architecture with a multi-headed self-attention mechanism. The encoder encodes the input sequence

and generates contextual representations using a self-attention mechanism to capture dependencies across the entire input sequence. These representations pass through a position-wise feed-forward neural network to introduce non-linearity and learn contextual representations. The decoder utilizes additional self-attention mechanisms to generate the output sequence based on the encoded contextual representations. The usage of multi-headed attention mechanism enables the Transformer model to perform multiple attention operations in parallel, efficiently capturing global and local dependencies from larger input sequences. The success of the Transformer model has influenced the development of various large-scale transformer-based pre-trained models based on encoder-decoder architectures.

2.2.1 BERT Framework

The popular Bidirectional Encoder Representation from Transformers (BERT) framework [5] introduces a state-of-the-art language representation model that is based on the Transformer auto-encoder architecture to capture deep bidirectional contextual representations from the input sequence and better understand the contextual relationship within a given text. As shown in Figure 2.3, the BERT framework introduces a two step process: pre-training BERT on large textual corpora and fine-tuning BERT on various downstream NLP tasks.

Pre-training BERT

The BERT model is pre-trained on large, unlabeled textual corpora that include publicly available text from the BooksCorpus dataset and English Wikipedia. These corpora provide a diverse range of general domain knowledge text to capture rich language representations. BERT introduces two unsupervised pre-training strategies, namely Masked Language Modeling (MLM) and Next Sentence Prediction (NSP).

In the MLM strategy, 15% of the input sequence are randomly masked, with 80% replaced

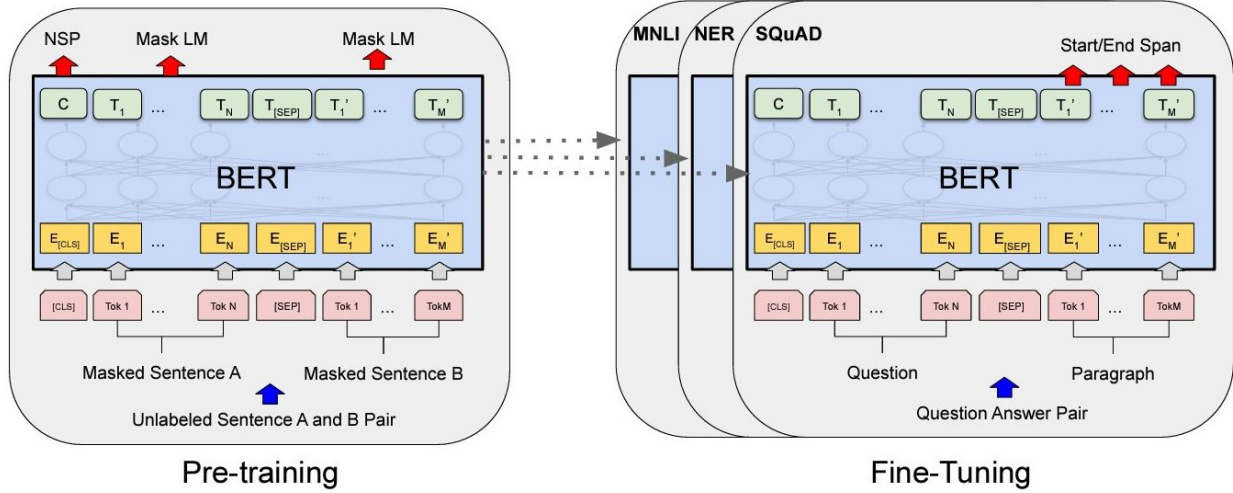


Figure 2.3: BERT Framework [5].

by the special "[MASK]" token, 10% replaced by a random token from the vocabulary, and the remaining 10% left unchanged. The model is trained to predict the original tokens from the masked ones. In the NSP strategy, pairs of consecutive input sequences (sentences) are labeled as "IsNext" or "NotNext" by replacing the second sentence with a random sentence 50% of the time. The model is trained to predict whether the second sentence follows the first sentence or not.

Fine-tuning BERT

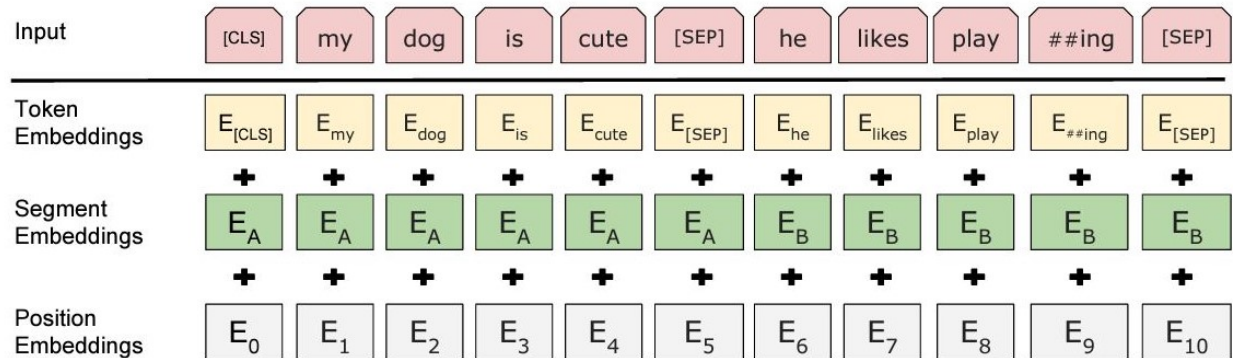


Figure 2.4: BERT Embeddings [5].

The pre-trained BERT model architecture for fine-tuning consists of three components,

namely an embedding layer, multiple Transformer Encoder Blocks, and task specific layers. In the embedding layer, BERT uses a pre-trained word-piece tokenizer to convert the input sequence into dense vector representations called word embeddings.

As shown in Figure 2.4, the embedding layer comprises three components, namely token embedding, segment embedding, and position embedding. Token embeddings map the pre-trained contextual representation to each word-piece token of the input sequence, segment embeddings distinguish between different segments in the input sequence using the special "[SEP]" token, and position embeddings encode positional information to differentiate word-pieces with the same token representation. The embedding layer outputs the summation of the token embeddings, segment embeddings, and position embeddings.

Each Transformer encoder block consists of a multi-head self-attention mechanism and feed-forward neural network layers. These components compute contextual representations from the input sequence and learn bidirectional contextual dependencies, capturing both local and global contextual information.

The task specific layers, often comprising fully-connected layers, are a crucial component that configures the BERT architecture to adapt to various downstream NLP tasks, such as Question Answering and Named Entity Recognition. The pre-trained BERT model can be fine-tuned by modifying both the input sequence and output formats to suit the requirements of the specific task during training.

2.2.2 BERT Variants

The BERT framework, which proposes a two-step approach for pre-training and fine-tuning large-scale Transformer-based models, has inspired the development of various BERT architecture variations and adaptations, such as RoBERTa [52], DistilBERT [53], ALBERT [54], ELECTRA [55], and T5 [56]. These BERT-based Transformer language models aim to

improve the state-of-the-art performance of the original BERT model and cater to specific NLP tasks [57, 8] or domain adaptations [58, 59].

Liu et al. proposed RoBERTa [52], which involves pre-training the BERT Transformer architecture on longer input sequences from larger and more diverse textual corpora, consisting of both private and publicly available resources. The masking positions of the training input sequences are dynamically changed for the MLM pre-training objective and the NSP pre-training objective is removed. Sanh et al. proposed DistilBERT [53], a smaller and lightweight variant of BERT, pre-trained using knowledge distillation to reduce computational costs while preserving the performance of larger models. Knowledge distillation introduces a teacher-student framework, where a smaller student model is trained to replicate the behavior of larger BERT models, acting as the teacher. The student model learns to approximate soft target probabilities from the teacher model. Yang et al. proposed the XLNet framework [4], which integrates an auto-regressive Transformer-XL language model. XLNet combines both auto-regressive and auto-encoding language modeling methods through the permutation language modeling pre-training objective, predicting the probability distribution over permutations of the input sequence.

Additionally, domain specific variants of BERT, such as SciBERT [60], BioBERT [58], and ClinicalBERT [61, 62, 63], leverage the versatility of the BERT framework to extend the application of the general-purposed BERT model to specialized domains. SciBERT [60] is pre-trained on scientific literature, BioBERT [58] on biomedical literature, and ClinicalBERT [61] on clinical documents, enabling them to capture domain specific language patterns as "prior knowledge" and enhance the performance on specialized tasks.

2.3 Domain Specific NER

Although NER approaches are known to push the state-of-the-art boundaries for entity extraction tasks in the general domain [16], specialized domains, such as biomedical [64, 65, 66, 67] and clinical [68, 69, 70, 71] domains, have also pose well-established challenges in this area of research.

The biomedical domain has been widely researched for domain specific NER (BioNER) tasks, presenting challenges, such as GENIA [64], JNLPBA [66], NCBI [65], and BC5CDR [67] for specialized entity extraction, including diseases, chemicals, proteins, and genes. Song et al. [72] introduced a hybrid dictionary-based approach. It combines fuzzy dictionary matching, employing a dictionary constructed from MeSH tree [73], with the shortest path edit distance (SPED) algorithm to approximate entity matches. They also apply POS-tagging rules for context-enabled text mining. Nayel et al. [14] proposed a biLSTM-CRF architecture that leverages information from the publicly available Merged Disease Vocabulary (MEDIC) dictionary [74] and biomedical domain pre-trained embeddings to enhance the contextual representations of the input tokens for entity classification. Jeong and Kang [38] introduced the ConNER approach that utilizes fully connected Multi-layer Perceptron (MLP) layers for the main classification task using contextual representations. It employs a biLSTM architecture to enhance label dependencies for document-level entity extraction.

The clinical domain also presents well-established challenges, such as I2B2 [68, 69], N2C2 [70], and MIMIC II [71], for the extraction of specialized named entities, including adverse drug events (ADE), drugs, and dosages from clinical documents. Wu et al. [75] conducted a study comparing the use of an RNN-based and a CNN-based deep learning architecture to extract clinical related concepts from clinical texts. The results demonstrate that the RNN-based LSTM-CRF model exhibits the best performance. Magge et al. [76] adopted a character-level BiLSTM-CRF model to perform named entity extraction for relationship

extraction using a Random Forest classifier in an NLP pipeline. Nayel et al. [77] proposed an ensemble framework that utilizes SVM models as base classifiers for different segment representations (e.g., BIO, BIOE, BIOES, and etc.), and the results are combined using majority voting and stacking techniques. Subsequently, a CRF-based meta classifier is used for sequence tagging.

Named entity recognition for the food, dietary, and nutrition domain has also gained attention in recent years [78, 79, 80, 81]. Popovski et al. [82] introduced FoodIE, which is a rule-based approach followed by an NLP pipeline that includes text pre-processing, POS-tagging, lemmatization, semantic tagging using conditional rules, and rule-based NER tagging of food entity chunks. Cenikj et al. [39] proposed BuTTER, a biLSTM-CRF architecture that leverages pre-trained word representation models, such as GloVe [40], Word2Vec [41], and FastText [42], to enhance the contextual representation of tokens for food entity classification. Perera et al. [13] proposed FooDCoNER, a hybrid model that combines a CRF model with a dictionary-based approach for feature extraction.

2.4 Low-resource Domain NER

Conventional machine learning and deep neural approaches for NER generally rely on large training datasets to achieve good performance in supervised settings. However, data scarcity issues caused by limited availability of annotated data often result in poor performance [8]. Low-resource problems are commonly encountered in specialized domains where large training datasets are usually unavailable [15, 57]. Additionally, the data annotation process is labour intensive, time-consuming, and heavily dependent on domain knowledge [8, 83].

2.4.1 Few-shot Learning

The zero-shot or few-shot NER tasks present challenges for entity extraction [84, 83, 85] due to limited training data. Lightweight approaches that utilize external resources, such as dictionaries [72, 14, 13, 48] and pre-trained text embeddings [74, 39], have been widely adopted for low-resource domain NER.

Nguyen et al. introduced the ZERO model, which leverages contextual representations from LUKE embeddings [86] and external word representations from GloVe [40] and ConceptNet [87] for zero-shot and few-shot NER. Das et al. [88] proposed CONTaiNER, a contrastive learning-based framework that models Gaussian embeddings and optimizes distributional divergence between token representations, enabling fine-tuning on small samples without the risk of over-fitting. Cui et al. [89] proposed a template-based approach that utilizes the BART language model [90] to generate templates from the input sequences. The approach exploits the generative capabilities of pre-trained encoder-decoder models to construct several templates and populate them with the candidate entity spans and entity types in a sequence-to-sequence generation manner.

2.4.2 Domain Adaptation

In recent advancements, large-scale Transformer-based language models, such as BERT [5], RoBERTa [52], and GPT [91], that are pre-trained on large general domain corpora have demonstrated effective performance in low-resource settings. However, the difference in word distribution between the pre-training corpora, such as Wikipedia and BookCorpus [5], and the target task can impact the performance of these pre-trained language models when adapting to specific domains [8, 15]. Consequently, recent works have adopted domain specialization [58, 61] and cross-domain adaptation approaches [57, 15] to enhance the generalization abilities of these models for low-resource domain NER tasks.

Lee et al. [58] introduced BioBERT by pre-training BERT [5] on large biomedical corpora using the same pre-training objectives proposed by BERT, and leveraging the pre-train weights from BERT without training the large BERT architecture from scratch. The pre-trained BioBERT model is then fine-tuned on various tasks, including Named Entity Recognition, Relationship Extraction, and Question-Answering, using multiple benchmark datasets from the biomedical and clinical domains. Jia et al. [15] constructed a cross-domain language model utilizing a parameter generation network that leverages contrasting embeddings from source and target domains to facilitate cross-domain knowledge transfer and enable supervised and unsupervised domain adaptation. Liu et al. [57] proposed a cross-domain BERT-based NER model that employs multiple domain related corpora for domain adaptive pre-training (DAP) [92] to further pre-train the BERT language model [5]. A human-annotated CrossNER dataset that covers five diverse domains with specialized entity categories is then used to fine-tune the cross-domain NER model. In [8], Liu et al. utilized Wikipedia anchors and DBpedia Ontology to construct a large-scale pre-training corpus of relatively high quality, which contains an abundance of entity categories. This corpus is used to pre-train an NER-BERT model, based on the BERT architecture [5], to enhance cross-domain adaptation and cross-domain NER fine-tuning.

2.5 Summary

In this chapter, we have reviewed the related work on named entity extraction. We discuss various approaches, including heuristic rule-based methods, feature-based machine learning models, deep neural models, and hybrid approaches, for the NER task. Specifically, we focus on Transformer-based architectures and frameworks, such as the pre-trained BERT language model [5], as well as hybrid NER approaches that combine deep neural models with heuristic rule-based or dictionary-based approaches. Domain specific NER tasks often encounter the low-resource problem due to the limited availability of training datasets. To

address this issue, zero-shot or few-shot NER approaches have been proposed as effective solutions for domain adaptation. With the emergence of large-scale pre-trained Transformer-based frameworks, domain adaptive pre-training strategies [92] and cross-domain adaptation transfer learning techniques [57] have demonstrated improved generalization abilities of language models for low-resource domains.

Chapter 3

Heuristic Entity Extraction (HEE)

As discussed in Chapter 1.4, the proposed NEE-FSM framework for food safety events monitoring consists of three major modules. Firstly, the Heuristic Entity Extraction (HEE) module curates the base dictionaries. Then, the Named Entity Recognition (NER) module augments the base dictionaries with more named entities. Finally, the Named Entity Matching (NEM) module extracts food safety entities for food safety events monitoring. In this chapter, we discuss the Heuristic Entity Extraction module of the NEE-FSM framework to curate the base dictionaries.

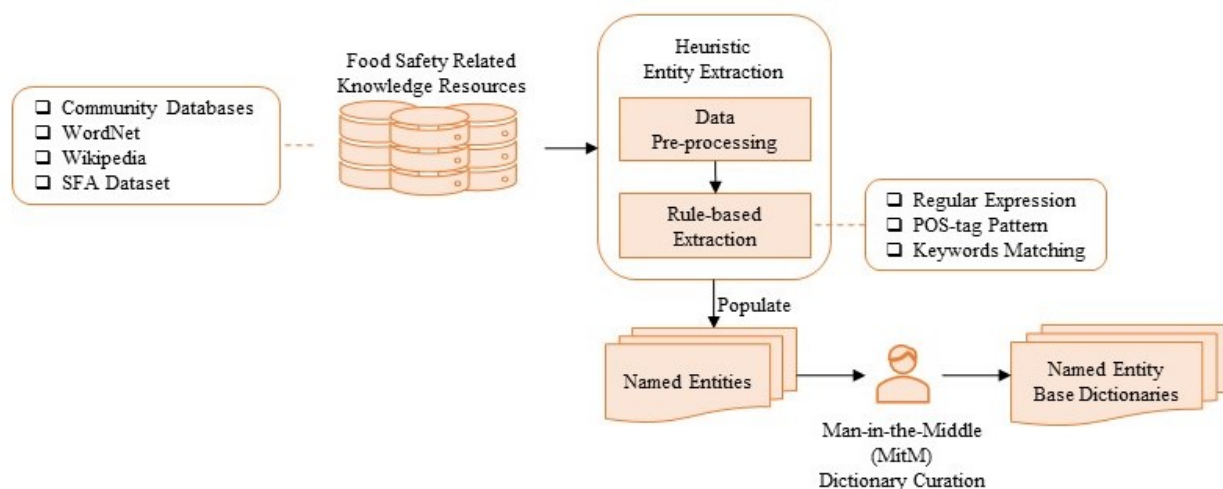


Figure 3.1: Proposed method for Heuristic Entity Extraction.

Figure 3.1 shows the proposed method for constructing the base set of dictionaries by extracting the food safety related entity mentions from various food safety related knowledge resources. These knowledge resources include the WordNet lexical taxonomy [93], Wikipedia (Wiki-pages), and several publicly available databases such as FoodData Central [94] and FooDB databases [95], that are collated by experts from various research communities. They contain comprehensive lists of commonly known entity terms for the food safety categories of interest. Then, data pre-processing and different rule-based approaches are applied to extract the food safety related named entities from the different knowledge resources. Finally, the extracted named entities are curated and verified by the SFA scientist to ensure the quality of the named entities for each food safety category.

To curate the base dictionaries, we extract the named entities by applying different rule-based heuristics with Regular Expressions (RegEx) and POS-tagging sequences for pattern matching, and keyword-based matching, to extract the specific lists of food safety related entity mentions from the different knowledge resources. The knowledge resources have a variety of data structures that determine the complexity of the extraction process. For example, the well-formatted community databases contain entity mentions that could be directly extracted. In contrast, the semi-structured Wiki-pages contain free-formed texts which require the use of more complex extraction techniques, such as pattern-based methods, for entity extraction.

In this research, we focus on the extraction of the named entities from four categories, including "*Disease*", "*Food*", "*Location*", and "*Symptom*" for the construction of the base dictionaries. In the following sections, we discuss the various heuristic approaches to extract the food safety entity mentions from the different knowledge resources to construct our base dictionaries for each entity category.

3.1 Food Entity Dictionaries

The essential information on common food products, such as food names, scientific names, constituent categories, food nutrients, descriptions, etc., are widely available in an abundance of knowledge resources collated by various research communities. These resources are typically used for food, nutrient, and dietary research studies [13]. Therefore, in our research, we focus on extracting food entities to construct the food entity dictionaries. Following [13, 96], we construct our food entity base dictionaries to capture comprehensive lists of commonly known food product mentions from the following three community food databases: FDC Database [94], FooDB Database [95], and FoodBase Corpus [79].

3.1.1 FoodData Central (FDC) Database

The FDC Database [94] is a public database managed by the USDA Agricultural Research Service, that provides downloadable information on food and nutrient across various categories for agricultural and experimental research. These categories include foundation food from commodity derived processed foods (e.g., whole grains, fresh food and vegetables, fresh or frozen meat, dairy products, etc.), branded food from international food labels, SR legacy food from historical food components from scientific literature, and survey food from the National Health and Nutrition Examination Survey (NHANES) for dietary studies.

To construct the base dictionaries, we extract the food entities from the foundation food and survey food category datasets. The downloaded datasets are well-structured in the JSON format, as shown in Figure 3.2. We apply key-value matching to extract the food description text using the *"description"* key (e.g., *{ "description": "Egg, whole, raw, frozen, pasteurized" }*). The extracted text is then split by a *","* delimiter (e.g., *["egg", "whole", "raw", "frozen", "pasteurized"]*) to extract the food entity *"egg"* from the first position. Then, the food entities are curated into a list, and the overlapping food entities are removed.

▼ FoundationFoods:		▼ SurveyFoods:	
▶ 0:	{-}	▶ 0:	{-}
▶ 1:	{-}	▼ 1:	
▶ 2:	{-}	foodClass:	Food
▶ 3:	{-}	description:	"Milk, NFS"
▶ 4:	{-}	▶ foodNutrients:	{-}
▶ 5:	{-}	▶ foodAttributes:	{-}
▼ 6:		foodCode:	"11100000"
foodClass:	"FinalFood"	startDate:	"1/1/2017"
description:	"Egg, whole, raw, frozen, pasteurized"	endDate:	"12/31/2018"
▶ foodNutrients:		▶ wwiaFoodCategory:	{-}
foodAttributes:	Food	fdcId:	1097511
▶ nutrientConversionFactors:	{-}	dataType:	"Survey (FNDDS)"
isHistoricalReference:	false	publicationDate:	"10/30/2020"
ndbNumber:	1171	▶ inputFoods:	{-}
fdcId:	323604	▶ foodPortions:	{-}
dataType:	"Foundation"	▶ 2:	{-}
▶ foodCategory:	{-}	▶ 3:	{-}
▶ foodPortions:	{-}	▶ 4:	{-}
▶ inputFoods:	{-}	▶ 5:	{-}
publicationDate:	"4/1/2019"	▶ 6:	{-}

Figure 3.2: Sample data from FoodData Central.

In total, we have extracted 1,858 food names.

3.1.2 Food Database (FooDB)

The FooDB Database [95] is a publicly available resource on food and dietary constituents. The FooDB Database includes comprehensive information on food constituents across several categories, such as food compositions involving food and its flavor, color, taste, aroma, etc., biochemistry, and physiology, which are collated from scientific literature.

To construct the base dictionary, we extract the food entities from the food category dataset. The food data is well-structured in JSON format as shown in Figure 3.3. We apply key-value matching to directly extract the food names using the "name" key (e.g., {"name": "Savoy cabbage"} to extract the food entity "savoy cabbage"). Then, the food entities are curated into a list and the overlapping food entities are removed. In total, we have extracted

▶ 0:	{-}
▼ 1:	Food Savoy cabbage Brassica oleracea var. sabauda Savoy cabbage (Brassica _um, iron and magnesium." itis_id: null wikipedia_id: "Savoy cabbage" picture_file_name: "2.jpg" picture_content_type: "image/jpeg" picture_file_size: 155178 picture_updated_at: "2012-04-20T09:39:54.000Z" legacy_id: 2 food_group: "Vegetables" food_subgroup: "Cabbages" food_type: "Type 1" created_at: "2011-02-09T00:37:15.000Z" updated_at: "2019-05-14T18:04:13.000Z" creator_id: null updater_id: null export_to_afcdb: false category: "specific" ncbi_taxonomy_id: 1216010 export_to_foodb: true public_id: "FOOD000002"
▶ 2:	{-}
▶ 3:	{-}

Figure 3.3: Sample data from FooDB.

1,014 food names.

3.1.3 FoodBase Corpus

The FoodBase Corpus [79] is a benchmark corpus that is annotated for the food entity extraction task, containing a comprehensive collection of labeled food entities from Allrecipes¹. The FoodBase Corpus consists of two datasets, a curated set of 1,000 recipes that are manually annotated with gold-standard labels, and an uncurated set of 21,699 recipes that are

¹Allrecipes ("https://www.allrecipes.com/") is a community platform that provides a range of recipes, cooking tips, and food-related contents

heuristically annotated [79]. The datasets are labeled in a hierarchical structure with 228 semantic tags from Hansard corpus, which are propagated from three root classes, including "Food & Drinks (AG)", "Animals (AE)" and "Plants (AF)". For example, "Food & Drinks (AG)" class has sub-categories that include "Food (AG:01)", "Production of food, farming (AG:02)" and "Acquisition of animals for food, hunting (AG:03)" classes.

```

<document>
  <id>0recipe106</id>
  <infon key="category">Appetizers and snacks</infon>
  <infon key="full_text">
    Preheat oven to 350 degrees F (175 degrees C). In a medium bowl combine the shredded chicken, picante sauce and ground cumin. Mix together. In a small skillet heat vegetable oil over medium high heat. Place a corn tortilla in the oil for 1 to 2 seconds on either side to soften. Repeat with all of the tortillas (if possible, have someone help you do this). Place tortillas on paper towel to soak up some of the oil. Put 1 tablespoon of the chicken mixture in the center of a tortilla and sprinkle some cheese on top. Roll up tortilla and place on a lightly greased cookie sheet, seam side down. Repeat with all of the tortillas until chicken mixture is gone. If there is any cheese left over, sprinkle it over the rolled tortillas. Bake tortillas in the preheated oven for 15 to 25 minutes or until tortillas are a little crispy and golden brown.
  </infon>
  <annotation i
    <location onse... length="16"/>
    <text>shredded chicken</text>
    <infon key="semantic_tags"> AG.01.d.06 [Fowls];</infon>
  </annotation>
  <annotation i
    <location onse... length="13"/>
    <text>picante sauce</text>
    <infon key="semantic_tags">
      AG.01.l.04 [Sauce/dressing];AG.01.h [Fruit and vegetables];
    </infon>
  </annotation>
  <annotation i
    <location onse... length="12"/>
    <text>ground cumin</text>
    <infon key="semantic_tags"> AG.01 [Food];</infon>
  </annotation>
  <annotation i
    <location onse... length="13"/>
    <text>vegetable oil</text>
    <infon key="semantic_tags"> AG.01.f [Fat/oil];</infon>
  </annotation>
  <annotation i
    <location onse... length="13"/>
    <text>corn tortilla</text>
    <infon key="semantic_tags">
      AG.01.i [Corn/cereals/grain];AG.01.n.11 [Bread];AG.01.n.12 [Pancake/tortilla/oatcake];
    </infon>
  </annotation>

```

Figure 3.4: Sample data from FoodBase.

To construct the base dictionary, we extract the food entities labeled with semantic tags from the well-structured datasets in XML format, as shown in Figure 3.4. First, we apply key-value matching to extract both food entity text and semantic tags using the "text" and "semantic_tag" keys respectively. We utilize the semantic tags to filter the food entities with the "AG" series of tags. The semantic tags are first split using a ";" delimiter, for food entities with multiple semantic tags (e.g., "picante sauce" contains "AG.01.l.04 (Sauce/dressing)" and "AG.01.h (Fruit and vegetables)"). Then, we apply Regular Expression (RegEx) rules to remove food entities with the "AE" and "AF" series of semantic tags. In addition, we also remove food entities that are only tagged with certain "AG" tags that

are non-food product related (e.g., "AG.01.s (Providing/receiving food)" and "AG.01.x.01 (Eating)"). Then, we perform post-processing, utilizing regular expression patterns and keywords matching to remove words from the food entity text (e.g., removing "additional" from "additional nutmeg"). Finally, the food entities are curated and manually verified to ensure the quality of extraction. In total, we have extracted 3,106 food names.

3.1.4 Singapore Cuisine Wiki-Page

In addition, to capture the local food names in Singapore, we also construct a dictionary for Singapore food entities from the "Singapore Cuisine" Wiki-page². We employ the heuristic rule-based approaches to extract food entities from well-structured lists, and from unstructured sentences of specific sections, such as introductory paragraphs and food item descriptions in the Wiki-Page. As can be seen in Figure 3.5, valuable food entities could also be found in the text descriptions.

Malay/Indonesian [edit]

Situated between Malaysia and Indonesia, Singaporean Malay dishes are influenced by the food of the neighbouring Malay Peninsula, Sumatra, Java and the Riau Islands. Despite absorbing regional influences, it tends to be adapted to local tastes and differs from their counterparts in neighbouring countries. Although Malays, such as the Orang Laut, are native to Singapore, most Malays in Singapore today are descended from native Indonesians or native Malays from present-day Malaysia.^[28] Hence, Singaporean Malay cuisine features a unique set of influences, especially from Minang cuisine. Spices and coconut milk are common ingredients, although Chinese ingredients such as **taupok (tofu puffs)** and **tofu** (known as **tauhu** in Malay) have been integrated. Many Chinese and Tamil Muslim adaptations of the following dishes also exist. As almost all Malays are Muslims, **pork** is not used as it is prohibited in Islam.

- **Acar**, pickled **vegetables** or **fruits** with **dried chilli**, **peanuts**, and **spices**. Indian and Peranakan versions can also be found.
- **Assam pedas**, **seafood** and **vegetables** cooked in a **sauce** consisting of **tamarind**, **coconut milk**, **chilli**, and **spices**.
- **Ayam penyet**, **fried chicken** dish consisting of **fried chicken** that is smashed with the **pestle** against mortar to make it softer and a relatively new culinary phenomenon of Indonesian origin.^[29]
- **Bakso**, also **Ba'so**, **meatballs** served with **noodles**.
- **Begedil**, **mashed potato mixture** that is fried into **patties** and eaten together with **mee soto**.
- **Curry puff**, also known as **epok-epok**, a **flaky pastry** usually stuffed with **curry chicken**, **potato cubes**, and a slice of **hard-boiled egg**. **Sardines** are sometimes used in place of **chicken**.



Figure 3.5: Sample data from "Singapore Cuisine" Wiki-page.

²https://en.m.wikipedia.org/wiki/Singaporean_cuisine.

First, we select the Wiki-page sections texts based on the header keywords, which include *"Chinese"*, *"Malay"*, *"Indian"*, *"Cross-cultural"*, *"Seafood"*, *"Fruit"*, *"Desserts"*, *"Ice cream sandwich"*, *"Drinks and beverages"*, and *"Singaporean dishes uncommon in Singapore"*. Then, we process each line of texts in each section in a NLP pipeline using the NLTK and SpaCy python packages. The sentences are firstly tokenized into a word-level sequence (e.g., *"Bakso, also Ba'so, meatballs served with noodles."* is split into [*"Bakso"*, *"."*, *"also"*, *"Ba'so"*, *"."*, *"meatballs"*, *"served"*, *"with"*, *"noodles"*, *"."*]). It is then followed by applying POS-tagging on the sequence of tokens (e.g., [*"NNP"*, *"."*, *"RB"*, *"NNP"*, *"."*, *"VBZ"*, *"VBN"*, *"IN"*, *"NNS"*, *"."*]), which is shown in Figure 3.6.

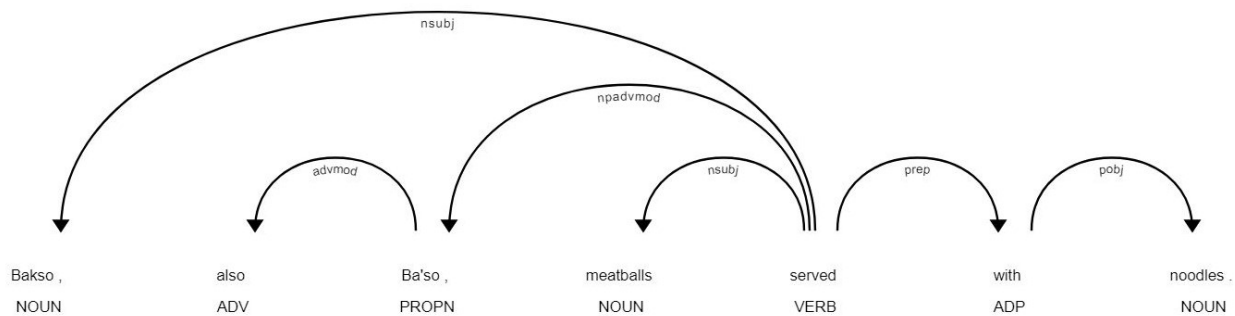


Figure 3.6: Sample of POS tagging sequence.

Then, utilizing a Regular Expression Parser, we extract food entity chunks by merging the POS-tags with a manually customized set of grammar (rules) (e.g., combining *"Adjective (JJ)"*, *"Nouns (NNS)"* and *"Verbs (VBN)"* instead of using only *"Nouns (NNS)"*), to account for the multi-lingual food names in Singapore. Since food names are often the direct conversion of different language pronunciations (e.g., Chinese, Malay, Tamil, etc.), to English spellings (e.g., *"niú ròu guǒ tiáo"*), where POS tagging sequences can vary across different languages (e.g., [*"JJ"*, *"NN"*, *"NN"*, *"NN"*]). Then, we perform post-processing by removing stop-words, punctuation, and numerical values from the entity chunks. In addition, we apply keyword matching to remove prefixes that denote the type of food (e.g., removing *"Singaporean"* from *"Singaporean thick rice noodles"* to obtain *"thick rice noodles"*). Finally, the food entities are curated and manually verified to ensure the quality of

extraction. In total, we have extracted 794 English food names.

3.2 Disease and Symptom Entity Dictionaries

The implicated food-borne related and zoonotic disease names, and the clinical symptom names are highly specialized named entities in the food safety domain. Therefore, to aid our extraction task, the SFA scientists have manually curated a list of disease and symptoms names that are specifically relevant for food safety monitoring. As shown in Table 3.1, the bio-surveillance dataset contains a list of food-borne diseases that are contracted through the human consumption of food items, which include pathogen, disease, and symptom names, and useful keywords. And a list of zoonotic diseases that impact food supply and production, which includes common disease names, affected animals, and useful keywords. These useful information are structured in a CSV format for direct extraction. The initial dataset consists of 32 food-borne disease names and 44 zoonotic disease names, followed by an additional set of 83 food-borne disease names and 21 zoonotic disease names. The symptom names are extracted and split by a “,” delimiter. In total, we have extracted 138 unique disease names, and 46 symptom names.

Disease	Category	Example
Food-borne Disease	Pathogen	hepatitis a virus
	Disease	hepatitis a
	Symptom	jaundice; fever; nausea; vomiting; abdominal pain; tenderness
	Keywords	contaminated; faecal-oral; undercooked; raw; food-handling
Zoonotic Disease	Disease	bovine tuberculosis
	Affected Animal	cattle; cow
	Keywords	zoonotic; mycobacterium bovis tb; contaminated; milk;

Table 3.1: SFA bio-surveillance dataset.

To expand the list of common symptom names, we leverage the comprehensive informa-

tion from the conceptual hierarchical structures of the WordNet lexical database of English terms [93]. We recursively extract all lemma terms from the root Synsets of the *"Symptom"* keyword in a top-down hierarchical manner. To achieve this, we first extract the root Synset using the noun keyword *"Symptom"* (e.g., *"symptom.n.01"* and *"symptom.n.02"* Synsets). Then, using the root Synsets, we extract the lemma term *"symptom"* and its list of Hyponyms denoting *"is-a"* relationships, and its Member Holonyms denoting *"is-part-of"* relationships (e.g., [*"abscess.n.01"*, *"ague.n.02"*, *"albuminuria.n.01"*, *"alkalinuria.n.01"*, ..., *"wheeziness.n.01"*, *"withdrawal_symptom.n.01"*] are hyponyms of *"symptom.n.01"*). The process repeats recursively to retrieve the list of hyponyms and holonyms for each Synset from the WordNet taxonomy and extracts its lemma terms as symptom entities (e.g., *"abscessed_tooth.n.01"* is a hyponym of *"abscess.n.01"*). Finally, the entities are corrected by replacing the delimiter *"_"* with a space *" "* (e.g., *"withdrawal_symptom"* is corrected to *"withdrawal symptom"*). In total, we have extracted 325 symptom names.

In the experiments, we have extracted a list of 1,047 disease names from WordNet. However, through the verification process with the SFA scientist, most of these disease names are not related to the food safety domain. Therefore, this list of disease names are not included in the base dictionaries.

3.3 Location Entity Dictionaries

To identify the location names for food safety monitoring, we focus on the names of countries, regions, provinces, states, and places. In our research, we utilize the *Pycountry* Python package that contains the ISO database as the standard to extract the information on countries, which include the country name, country code, and official names (e.g., *"Country(alpha_2='AF', alpha_3='AFG', name='Afghanistan', numeric='004', official_name='Islamic Republic of Afghanistan')"*). Here, we utilize the keywords *"name"* and *"official_name"* to

extract the list of country names for the location entities. In total, we have extracted 453 country names. While our list of locations is only limited to a set of country names, we aim to expand this list through the NER module.

3.4 Performance Results

NE Category	Knowledge Resource	#Entities	#Total (Unique) Entities
Food	FoodData Central [94]	1867	5,866
	FoodDB [95]	1014	
	FoodBase [79]	3106	
	Wikipedia	794	
Disease	SFA Dataset	138	138
Symptom	SFA Dataset	46	328
	WordNet [93]	325	
Location	PyCountry Library	453	453

Table 3.2: Curated NEM base dictionaries.

Table 3.2 shows the number of unique named entities extracted from the various collection of knowledge resources for the base dictionaries. In total, we have extracted 6,785 commonly known food safety named entities for the specific categories in the food safety domain, which includes 138 *"Disease"* entities, 5,866 *"Food"* entities, 453 *"Location"* entities, and 328 *"Symptom"* entities.

3.5 Summary

In summary, this chapter has discussed the processes to curate the base dictionaries for food safety related named entities with relatively high quality, by leveraging the various publicly available knowledge resources, such as WordNet, Wikipedia, and community databases. In particular, we have extracted the comprehensive lists of commonly known named entities for four food safety categories, namely *"Food"*, *"Disease"*, *"Symptom"*, and *"Location"*.

We have discussed the different techniques used for the Heuristic Entity Extraction (HEE) module, such as the application of Regular Expression (RegEx) and POS-tagging sequences for pattern matching, and keyword-based matching, that are tailored specifically to the different knowledge resources for extracting the food safety related named entities.

Chapter 4

Named Entity Recognition (NER)

The Named Entity Recognition (NER) module aims to extract new and "unseen" food safety related entities to augment and enhance the coverage of the base dictionaries in the proposed NEE-FSM framework. It detects and classifies named entities from the target data, i.e., online food safety articles and customer feedback reports, for food safety categories, namely "*Disease*", "*Food*", "*Location*", and "*Symptom*".

Figure 4.1 shows the proposed approach for constructing a food safety domain specialized NER model under low-resource settings due to insufficient training data. The proposed approach consists of two main processes, namely Domain Adaptive Pre-training (DAP) and Task Specific Fine-tuning (TSF). In the DAP process, Pre-training Corpora Creation creates the unlabeled pre-training corpora by extracting textual contents from food and food safety related knowledge resources. Then, Pre-training performs the unsupervised domain adaptive pre-training [92] with the pre-trained BERT model [5] for food safety domain specialization. This process aims to continue pre-train the pre-trained BERT model [5] on the food safety related data to improve domain adaptive capabilities to capture more specialized food safety related entities.

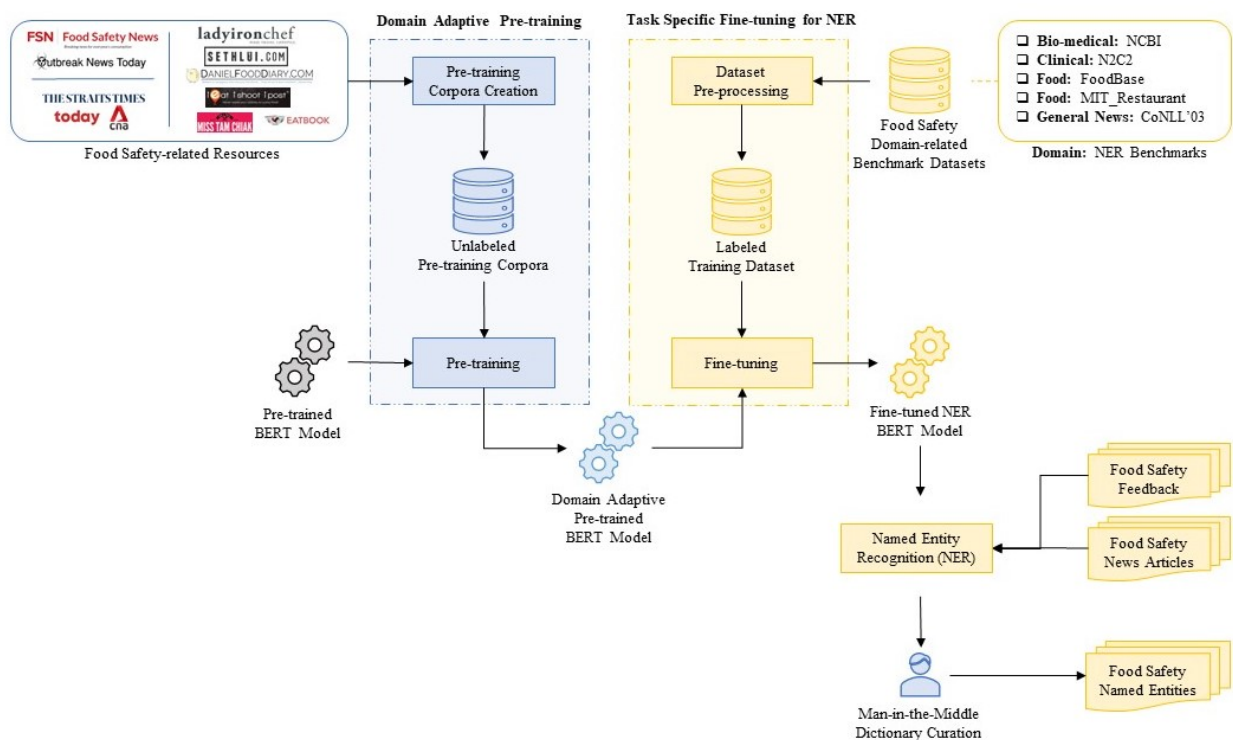


Figure 4.1: Proposed approach for Named Entity Recognition (NER).

In the TSF process, we fine-tune the pre-trained BERT model for the food safety NER task by leveraging multiple benchmark datasets from domains that are conceptually related to the food safety domain for transfer learning. We first prepare the benchmark datasets for the NER training. Dataset Pre-processing is then performed to filter the labeled data from the *"Disease"*, *"Food"*, *"Location"*, and *"Symptom"* categories in the food safety domain to build a labeled training dataset. Finally, Fine-tuning uses the labeled training dataset for fine-tuning the pre-trained BERT model using sequence labeling for the NER task. As a result, a NER fine-tuned model is generated.

Finally, the NER fine-tuned BERT model is applied to extract food safety related named entities from the food safety datasets to augment the base dictionaries. In addition, the extracted named entities are curated and verified by the SFA scientist to ensure the quality of the named entities for each food safety category.

4.1 Domain Adaptive Pre-training

The Transformer-based language models, such as BERT [5], that are pre-trained on large corpora containing general domain texts (e.g., Wikipedia and BookCorpus), have been shown to be effective for adapting to low-resource domains. However, the variance of word distributions between the pre-training corpus and target domain corpus may influence the pre-trained model’s ability to generalize on specialized domains [8]. Thus, domain specialization approaches that introduce ”prior knowledge” on the target domain by pre-training with target domain corpora have gained significant attention from the research community for facilitating domain adaptation [58, 61, 60, 97]. The Domain Adaptive Pre-training (DAP) process consists of two components, namely Pre-training Corpora Creation and Pre-training.

4.1.1 Pre-training Corpora Creation

As the target food safety datasets contain data from online food safety news articles and local customer feedback texts, we perform web-scraping on various food and food safety related resources to acquire an unlabeled textual corpora with specific contents and structures for the Domain Adaptive Pre-training process. For the extraction of web-based contents, individual web-scraping scripts are written to crawl the URLs from online resources. Using these URLs, we extract the main textual contents from specific HTML sections of the web-pages.

As discussed in [57], noisy and domain unrelated text found within a pre-training corpus may hinder the effectiveness of the pre-trained models for domain adaptation. Therefore, we aim to collate specialized corpora that contain different types of textual data for pre-training the BERT model [57]. We classify these data as the Task-level corpora and Domain-level corpora.

Task-level Corpora

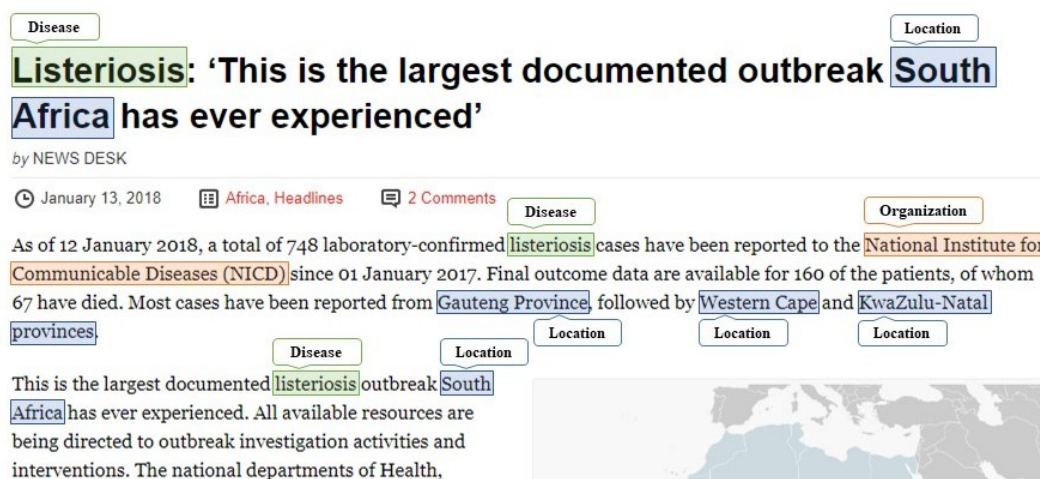


Figure 4.2: Example task-level content from Online Food Safety News Article.



Figure 4.3: Example task-level content from CRMS feedback Case Description.

The task-level corpora consist of textual contents that are explicitly related to the target food safety named entity extraction task. The corpora are created by extracting directly from the websites that contain food safety news articles, such as "Food Safety News" and "Outbreak News Today", and the customer feedback case descriptions from the CRMS reports. Figure 4.2 and Figure 4.3 show two examples on task-level contents. Pre-training on task-level corpora allows the model to directly learn and generalize the contextual information of the target data. For example, as shown in Figure 4.3, capturing structural contents from the CRMS reports can include information on localized terms or "slangs", food product names, and location names pertaining to Singapore. In total, we have extracted 71,801 articles with 983,598 sentences for the task-level corpora.

Domain-level Corpora

In this research, we introduce more textual contents from additional knowledge resources for the domain adaptive pre-training process. The domain-level corpora consist of textual contents that are related to the food and food safety domain categories of interest, such as "Disease", "Food", "Location", and "Symptom". The large domain-level corpora serve to provide more contextual coverage for domain adaptive pre-training that aims to capture more specialized food safety related terms in the target data. In this research, we acquire the contents from two types of online resources.

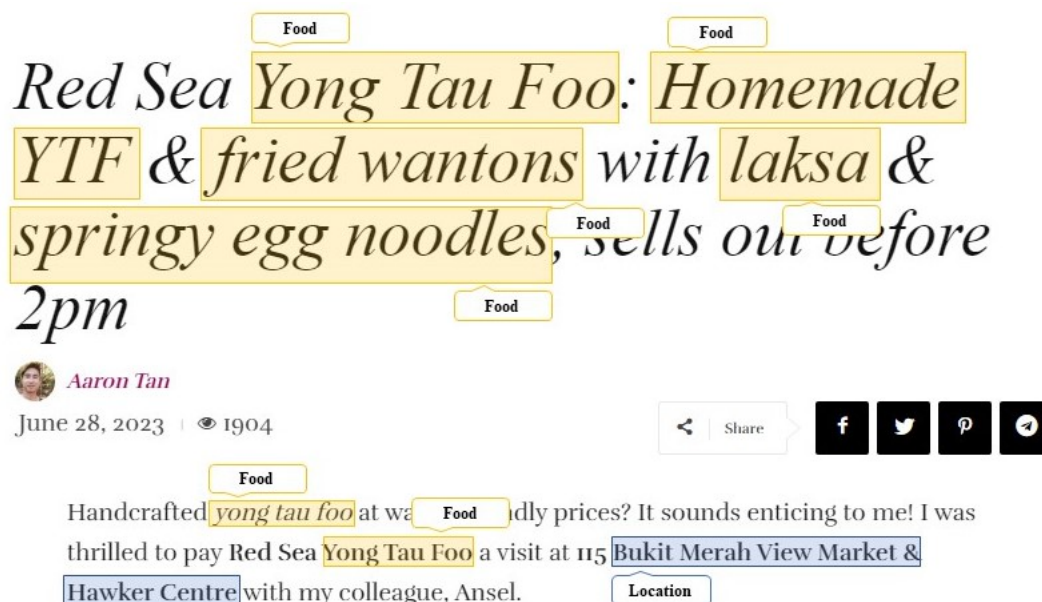


Figure 4.4: Example content from Singapore Food Blogs.

Firstly, we web-scrape and extract the contents from six popular Singapore food blogs, namely "DanielFoodDiary", "EatBook", "I Eat I Shoot I Post", "LadyIronChef", "Miss Tam Chiak", and "SEITHLUI.COM". As shown in Figure 4.4, these blog articles contain sentences with specific Singapore food items, such as "yong tau foo", "fried wantons", and "laksa", and location names, such as "Bukit Merah View Market & Hawker Centre". In addition, we also web-scrape and extract the article contents from three Singapore-based news providers, namely "Channel News Asia", "The Straits Times", and "Today Online".

Event Business Management, the operator of a temporary Ramadan fair, did not have its application approved to run the bazaar from April 9 to 24, 2022, the Singapore Food Agency said.

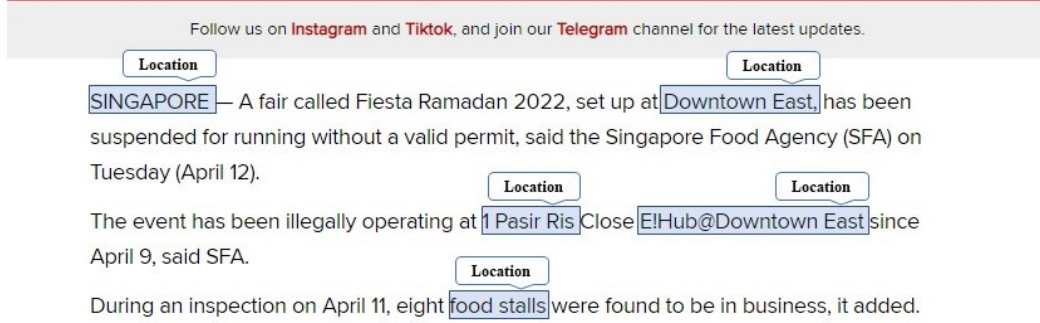


Figure 4.5: Example content from "Today Online" news website.

We utilize a list of keywords to search the contents of news articles that are related to food and food safety events. Figure 4.5 shows the news contents from "Today Online". In total, we have extracted 31,959 articles with 1,075,300 sentences for the domain-level corpora.

Pre-training Dataset

Corpus Type	Knowledge Resources	#Articles	#Total Articles	#Total Sentences
Task-level	<i>Food Safety News</i>	19,907	71,801	983,598
	<i>Outbreak News Today</i>	13,876		
	CRMS Dataset	38,018		
Domain-level	<i>DanielFoodDiary</i>	3,567	31,959	1,075,300
	<i>EatBook</i>	4,741		
	<i>I Eat I Shoot I Post</i>	1,790		
	<i>LadyIronChef</i>	4,347		
	<i>Miss Tam Chiak</i>	3,156		
	<i>SETHLUI.COM</i>	5,718		
	<i>Channel News Asia</i>	1,200		
	<i>The Straits Times</i>	3,142		
	<i>Today Online</i>	1,931		

Table 4.1: Domain adaptive pre-training corpora.

The Table 4.1 shows the statistics of unlabeled pre-training corpora based on the corpus type. We have web-scraped and extracted a total of 103,760 article contents with 2,058,898 unlabeled sentences. Among them, there are 71,801 articles with 983.598 sentences for Task-level Corpora, and 31,959 articles with 1,075,300 sentences for Domain-level Corpora. In our

experiments, we will investigate the effectiveness of domain adaptive pre-training by using different corpus types.

4.1.2 Pre-training

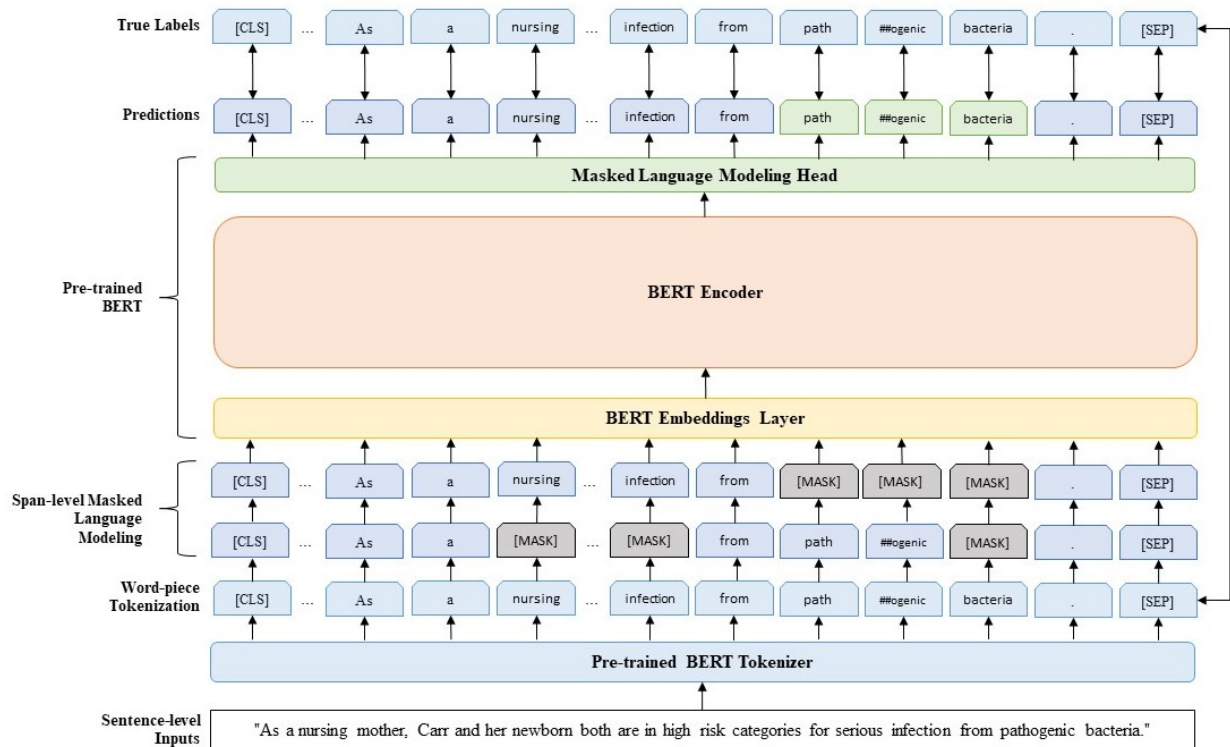


Figure 4.6: Domain adaptive pre-training of BERT Model.

Our work is inspired from Liu et al. [57] and other related works [58, 97] on the application of domain adaptive pre-training (DAP) strategy [92] to enhance the domain adaptive capabilities of the pre-trained BERT language model [5]. The proposed approach aims to specialize the pre-trained BERT language model [5] to capture more specialized terms for the food safety NER task by learning from the underlying contexts, lexicons and structural information specific to the food safety domain.

We perform DAP by leveraging the well-adopted framework for pre-training the BERT language model [5] on the unlabeled food safety related corpora. We utilize the BertFor-

MaskedLM pre-trained model packages from *HuggingFace* to pre-train the BERT model. Figure 4.6 shows the overall architecture of the domain adaptive pre-training process using the unsupervised Masked Language Modeling pre-training strategy. The model architecture consists of the following components for pre-training BERT: Pre-trained BERT Tokenizer, Span-level Masked Language Modeling, and Pre-trained BERT For Masked Language Modeling.

Pre-trained BERT Tokenizer

We apply the Pre-trained BERT Tokenizer (*BertTokenizerFast*) to encode the sentence-level input into word-piece tokens from the food safety pre-training corpora. The BERT Tokenizer generates a sequence of word-piece tokens, indexed with token IDs from the pre-trained BERT vocabulary. The special tokens "[CLS]" and "[SEP]" are added to denote the start and end of the sequence, respectively. Then, "[Pad]" tokens are added to truncate the uniform input sequence with maximum token length of 512.

Span-level Masked Language Modeling

We adopt the Masked Language Modeling (MLM) pre-training strategy introduced by BERT [5, 57] to learn the contextual representations of an input sequence by predicting the masked word-piece tokens. In addition, we also follow Liu et al. [57] to introduce a step to convert the random token-level masking into a span-level masking for the input sequence, as shown in Figure 4.6. To achieve this, we first apply random masking on the input word-piece sequence to flag out the tokens to be masked with a probability of 15%. Then, we shift the positions of masking flags to its adjacent positions into one another, towards the last flagged position to create a continuous masking span. For example, if the input sequence is ["As", "a", "nursing", ..., "infection", "from", "path", "##ogenic", "bacteria", "."],

and the tokens "nursing", "infection", and "bacteria" are randomly flagged for masking, the sequence becomes ["As", "a", [MASK], ..., [MASK], "from", "path", "##ogenic", [MASK], "."]. After that, the token-level masking positions are shifted towards the last masking position for the token "bacteria" to obtain the masking span as ["As", "a", "nursing", ..., "infection", "from", [MASK], [MASK], [MASK], "."]. Finally, out of these tokens that are flagged for masking, 80% are replaced by the "[MASK]" token, 20% are replaced with a random token from the pre-trained BERT vocabulary, and 20% remain unchanged. This is illustrated in Figure 4.6.

As the unsupervised masked language modeling approach utilizes the unlabeled corpora for pre-training BERT, the original input sequence of tokens is duplicated as the target (true) labels for the BERT model to predict the input token sequence from a masked input sequence. While the span-level masking strategy increases the complexity for MLM pre-training, such as learning to predict a longer sequence ["path", "##ogenic", "bacteria"] instead of random tokens such as "nursing", "infection", and "bacteria", it facilitates the pre-trained language model to better understand the specialized terms, structure and context of the target domain texts [57].

BERT for Masked Language Modeling

The masked input sequences are parsed into the *BERTForMaskedLM* model [5] which consists of the BERT Embedding Layer, BERT Encoder that consists of 12 Transformer encoder blocks (each encoder block with 12 attention heads, hidden size of 768, and 110 million parameters), and an additional Masked Language Modeling Head that consists of 2 linear layers for pre-training.

In this research, we initialize the pre-trained model and weights of the *bert-base-cased* model from *HuggingFace*. The vectors of token representations are extracted from input

sequence with the pre-trained BERT Embeddings. The token representations are parsed into the 12 BERT Encoder layers to learn the contextual representations between the tokens of the input sequence using the self-attention mechanism. Then, the contextual representation vectors are parsed into the Masked Language Modeling Head to predict the original sequence of word-piece tokens from the masked input sequence. To obtain the specialized pre-trained embeddings, the cross-entropy loss is calculated using the predicted sequence and the original input sequence after each pre-training iteration (epoch), for back propagation to update the original weights of the BERT pre-trained model.

4.1.3 Domain Adaptive Pre-trained BERT Models

Using the unlabeled pre-training corpora given in Table 4.1, we perform the Domain Adaptive Pre-training (DAP) on the pre-trained BERT model [5] to develop the Domain Adaptive Pre-trained BERT models, to adapt to the context of each target food safety dataset for the NER task.

Pre-trained Model	Corpus Type	Resource Description	#Articles	#Sentences
BERT [5]	None	None	None	None
FS-BERT	Task-level	Food Safety News Articles	65,742	1,809,310
	Domain-level	Singapore Food Blogs Food & Food Safety News Articles		
FS-BERT _{News}	Task-level	Food Safety News Articles	33,783	734,010
CRMS-BERT	Task-level	CRMS Dataset	38,018	249,588
CRMS-FS-BERT	Task-level	Food Safety News Articles CRMS Dataset	103,760	2,058,898
	Domain-level	Singapore Food Blogs Food & Food Safety News Articles		
CRMS-FS-BERT _{News}	Task-level	Food Safety News Articles CRMS Dataset	71,801	983,598

Table 4.2: Domain Adaptive Pre-trained BERT Models.

Table 4.2 shows the list of pre-trained models used for our experiments. In this research, we include the vanilla pre-trained BERT model [5] (without applying DAP) for comparison to evaluate the effectiveness of the domain adaptive pre-training (DAP) approach, as our proposed models are based on it:

- **BERT** [5] - It is a widely-adopted pre-trained model that uses a transformer architecture to learn contextual relationships from large corpora. The BERT framework enables the flexibility to fine-tune the pre-trained language model for various NLP tasks, including NER. In this research, we use the *bert-base-cased* model.

For the context of online food safety news articles, we have developed two variants of the food safety pre-trained BERT models as follows:

- **FS-BERT** - Pre-trained BERT model [5] that is also pre-trained on the articles from the online food safety news articles (i.e., "*Food Safety News*" and "*Outbreak News Today*"), Singapore food blogs (i.e., "*DanielFoodDiary*", "*EatBook*", "*I Eat I shoot I Post*", "*LadyIronChef*", "*Miss Tam Chiak*", "*SETHLUI.com*"), and the food and food safety related news articles (i.e., "*Channel News Asia*", "*The Straits Times*", "*Today Online*").
- **FS-BERT_{News}** - Pre-trained BERT model [5] that is also pre-trained on the task-level corpora that consist only of the online food safety news articles (i.e., "*Food Safety News*" and "*Outbreak News Today*").

Due to the difference in context, tonality, and structures of the sentences in the food safety news articles and customer feedback texts, we further perform DAP on the pre-trained BERT model, FS-BERT model and FS-BERT_{News} model on the CRMS dataset. This allows the pre-trained models to capture the context of the customer feedback texts to identify more specialized food safety related terms in Singapore. These pre-trained models are as follows:

- **CRMS-BERT** - Pre-trained BERT model [5] that is also pre-trained on the Task-level corpora that consist only of the CRMS feedback texts.
- **CRMS-FS-BERT** - Pre-trained FS-BERT model that is also pre-trained on the CRMS dataset. Therefore, it is pre-trained on the entire pre-training corpora.

- **CRMS-FS-BERT_{News}** - Pre-trained FS-BERT_{News} model that is also pre-trained on the CRMS dataset. Therefore, it is pre-trained on both task-level corpora from online food safety news articles and CRMS feedback texts.

4.2 Task Specific Fine-tuning

To address the low-resource domain NER issues due to insufficient training datasets, we adopt cross-domain adaptation methods [57, 8] to perform transfer learning for NER tasks. We fine-tune the pre-trained BERT model [5] using multiple benchmark datasets for the "*Biomedical*", "*Clinical*", "*Food*", and "*General*" domains, to construct the NER models to extract specific named entities for the target "*Food safety*" domain. Here, the source domains are conceptually related to the target domain that contains the food safety related categories such as "*Disease*", "*Symptom*", "*Food*", and "*Location*". The proposed approach aims to leverage the publicly available resources to alleviate the manual process for annotating the training datasets from scratch. The Task Specific Fine-tuning (TSF) process consists of the following components: Domain Specific Benchmark Datasets, Dataset Pre-processing, and Fine-tuning.

4.2.1 Domain Specific Benchmark Datasets

The domain specific benchmark datasets are sourced from various NLP research tasks on entity extraction (e.g., NCBI Disease corpus [65], National NLP Clinical Challenge (N2C2) [70], etc.). These publicly accessible benchmark datasets are largely well-annotated with "gold-standard" labels by various research communities, providing an abundance of specialized entity categories. As such, we aim to bypass laborious efforts and computational costs for labeling the target datasets from scratch.

In this research, we have identified four specialized domains from the NLP research task on entity extraction with publicly available training datasets that are conceptually related to *"Food Safety"*. The domains include the *"Biomedical"*, *"Clinical"*, *"Food"* and *"General"* domains. More specifically, we identify benchmark datasets that contain the subsets of the target food safety domain, i.e., *"Disease"*, *"Symptom"*, *"Food"*, and *"Location"* for NER modeling. In this section, we discuss the NER benchmark datasets that are used for fine-tuning the pre-trained BERT model for NER modeling.

Biomedical Domain Benchmark Dataset

- **NCBI (Disease)** [65] - The NCBI Disease corpus is well-established in the biomedical NLP research domain. It consists of 793 biomedical abstracts from the PubMed database and manually annotated with 6,892 disease mentions from the MeSH/OMIM concepts: *"Composite mention"*, *"Disease class"*, *"Modifier"* and *"Specific disease"*. For simplicity, these entities are commonly grouped together as the *"Disease"* category.

Clinical Domain Benchmark Dataset

- **N2C2** [70] - The National NLP Clinical (N2C2) Challenge comes from the popular series of I2B2 NLP challenges [68, 69]. The N2C2 challenge involves the extraction of medical concepts. The dataset consists of 505 medical reports from the MIMIC database containing specialized categories: *"Drug"*, *"Strength"*, *"Dosage"*, *"Duration"*, *"Frequency"*, *"Form"*, *"Route"*, *"Reason"* and *"Adverse Drug Event (ADE)"*, which focus on the extraction of clinical adverse drug events and attributes.

Food Domain Benchmark Dataset

- **FoodBase** [79] - The FoodBase Corpus consists of food recipes extracted from Allrecipes. In this research we utilize the curated set of 1,000 recipes with gold annotations. The food entities are annotated with multiple semantic tags from Hansard corpus that contain three root classes: "*Food & Drinks (AG)*", "*Animals (AE)*", and "*Plants (AF)*". We simplify the annotated labels by applying a set of heuristic rules to relabel each entities with the categories: "*Animal*", "*Food*", and "*Plant*".
- **MIT_Restaurant** [78] - The English-language MIT restaurant review dataset is extracted from online restaurant reviews that consist of summarised opinions of commuters via spoken dialogues. The dataset contains the following categories: "*Amenity*", "*Cuisine*", "*Location*", "*Restaurant Name*", "*Rating*", "*Hours*", "*Dish*" and "*Price*".

General Domain Benchmark Dataset

- **CONLL'03** [16] - The CONLL 2003 English dataset is a widely adopted benchmark dataset for NER research for general domain texts. It consists of 1,393 news articles from Reuters News for 4 generic categories: "*Person (PER)*", "*Organization (ORG)*", "*Location (LOC)*" and "*Miscellaneous (MISC)*".

4.2.2 Dataset Pre-processing

Dataset Pre-processing is performed to prepare the benchmark datasets for the target NER task specific fine-tuning. Table 4.3 shows the collection of benchmark datasets from domains that are conceptually related to the food safety domain, with their specialized category labels. The benchmark categories are filtered and modified to the target categories for the food safety domain.

Domain	Benchmark Dataset	Benchmark Categories	Target Categories
Biomedical	NCBI (Disease) [65]	Composite Mention, Disease Class, Modifier, Specific Disease	Disease
Clinical	N2C2 [70]	ADE , Dosage, Drug, Duration, Form, Frequency, Reason , Route, Strength	Symptom
Food	FoodBase [79]	Animal, Food , Plant	Food
	MIT_Restaurant [78]	Amenity, Cuisine, Location, Restaurant Name, Rating, Hours, Dish , Price	Food
General	CoNLL'03 [16]	Location , Miscellaneous, Organization, Person	Location

Table 4.3: Benchmark datasets with curated named entity categories of interest for NER task specific fine-tuning. Benchmark categories selected for target categories are in bold.

Named Entity Category Selection

In Dataset Pre-processing, we first perform named entity category selection from the categories available for each benchmark dataset. Then, we select the name entity categories of interest that are either directly relevant to the food safety NER task, such as "*Disease*", "*Food*", and "*Location*", or categories that are strongly related to the food safety categories of interest, such as "*Adverse Drug Event (ADE)*" $\rightarrow$ "*Symptom*", and "*Dish*" $\rightarrow$ "*Food*". The benchmark categories, such as "*Dosage*", "*Drug*", "*Duration*", and "*Form*", that are irrelevant to the food safety NER task are excluded from the selection process.

The category selection process relies on specific domain knowledge to identify the specific categories for the target food safety NER task. We have derived four target categories from the benchmark dataset, namely "*Disease*", "*Food*", '*Location*', and "*Symptoms*" from the respective domain specialized benchmarks, which is shown in Table 4.3.

Training Dataset Curation

We curate the datasets to facilitate NER training. We standardize all annotated labels from each dataset using the BIO-tagging scheme [98] for each sentence split. The BIO (Beginning, Inside, Outside) tagging scheme provides a common format for the computational linguistic task, such as NER. It tags token sequences into chunks by using the "B" and "I" prefixes to indicate the beginning and the inside of an entity chunk respectively (e.g., the word-level token of "South Africa" is represented as ["South", "Africa"] → ["B-Location", "I-Location"]). The "O" tag indicates that a token is not identified as a named entity. This creates pair-wise mappings for each input text sequence to its corresponding BIO-tag annotation sequence at the word-level. For example, ["The", "risk", "of", "cancer", ",", "especially", "lymphoid", "neoplasias", ..., "instability", "."] is annotated as ["O", "O", "O", "B-Disease", "O", "O", "B-Disease", "I-Disease" ..., "O", "O"] with the word "cancer" and word span ["lymphoid", "neoplasias"] are labeled as "Disease" entities.

Using the selected categories, we modify and re-label the annotations of the benchmark datasets. Given the following example from the N2C2 benchmark dataset [70] for the sentence "The pain was adequately controlled on 40mg of OxyContin t.i.d.", we perform the following steps for annotations:

- First, we extract the word-level token sequence ["The", "pain", "was", "adequately", "controlled", "on", "40", "mg", "of", "OxyContin", "t.i.d", "."] and its original labels provided by the benchmark dataset ["O", "B-Reason", "O", "O", "O", "O", "B-Strength", "I-Strength", "O", "B-Drug", "B-Frequency", "O"].
- Next, we remove the irrelevant categories such as "Strength", "Drug", and "Frequency" from the annotations to obtain the labeled sequence ["O", "B-Reason", "O", "O", "O", "O", "O", "O", "O", "O", "O"]

- Then, we rename the related categories to the target food safety categories, for example, "*ADE*" and "*Reason*" categories are renamed to "*Symptom*" to obtain the target labeled sequence ["*O*", "*B-Symptom*", "*O*", "*O*", "*O*", "*O*", "*O*", "*O*", "*O*", "*O*", "*O*", "*O*"] to fine-tune our NER model. This step is not required for the benchmark categories that are equivalent to the target food safety categories (e.g., "*Disease*", "*Food*", and "*Location*"), as shown in Table 4.3.
- Finally, as the category selection process introduces significant number of negative samples which may cause data imbalance in category classes for NER training, we remove the sentence and label pairs with annotated sequence that consist entirely of "*O*" labels.

Fine-tuning Dataset

Domain	Benchmark Dataset	#Total Sentences
Biomedical	NCBI (Disease) [65]	3,958
Clinical	N2C2 [70]	6,515
Food	FoodBase [79]	996
	MIT_Restaurant [78]	1,678
General	CoNLL03 [16]	7,722
Total		20,869

Table 4.4: Curated benchmark datasets for task specific fine-tuning.

Table 4.4 shows the statistics of the curated benchmark datasets with a total of 20,869 annotated data from the specialized domains, including "*Biomedical*", "*Clinical*", "*Food*", and "*General*" domains. In this research, we use the annotated benchmark data to fine-tune the domain adaptive pre-trained BERT models for food safety NER to extract named entities from the "*Disease*", "*Symptom*", "*Food*", and "*Location*" categories.

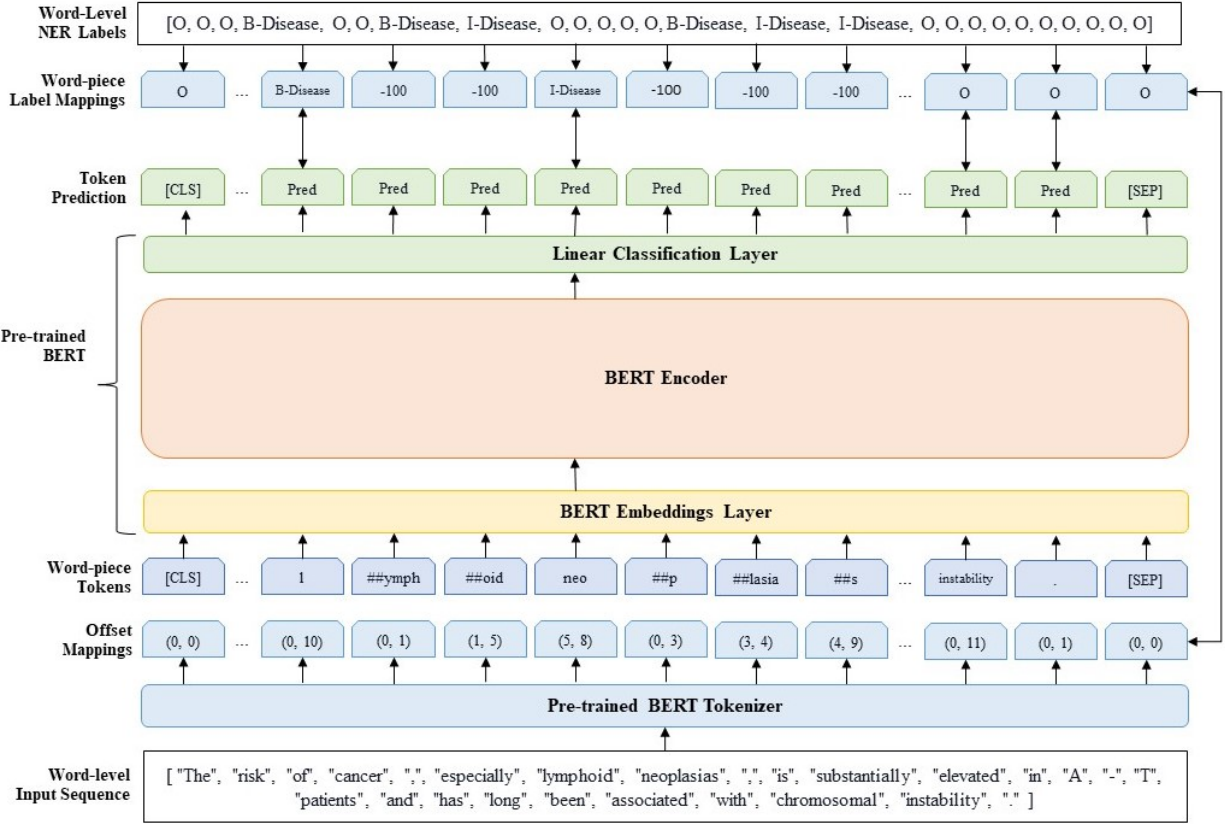


Figure 4.7: Task Specific Fine-tuning of Pre-trained BERT Model for NER task.

4.2.3 Fine-Tuning

We perform the NER Task Specific Fine-tuning on the domain adaptive pre-trained BERT models using the supervised learning approach for sequence tagging [5] to predict the sequence of NER labels from the word-level input sequence. We utilize the *BertforToken-Classification* pre-trained model packages from *HuggingFace* for NER modeling. Figure 4.7 illustrates the fine-tuning process of the pre-trained BERT model for a sequence-to-sequence classification. The model architecture consists of the following components for fine-tuning BERT: Pre-trained BERT Tokenizer and BERT for Token Classification.

Pre-trained BERT Tokenizer

We apply the Pre-trained BERT Tokenizer (*BertTokenizerFast*) to encode the input word-level input sequence into word-piece tokens. The BERT tokenizer generates a sequence of word-pieces tokens, indexed with token IDs from the pre-trained BERT vocabulary. Special tokens "[CLS]" and "[SEP]" are added to denote the start and end of input sequence respectively. The "[Pad]" tokens are added to truncate uniform input sequences with maximum token length of 512.

As a single word may be split into multiple word-pieces and the NER labels are in word-level sequence (e.g., ["lymphoid", "neoplasias"] → ["B-Disease", I-Disease]), we extract the offset mappings (positional information) to align each NER label to the first word-piece of the word token. We assign "-100" to the remaining word-pieces to ignore the offset tokens during training. For example ["lymphoid", "neoplasias"] is split into ["l", "##ymph", "##oid", "neo", "##p", "##lasia", "##s"] word-piece tokens with the word-piece label mappings of ["B-Disease", -100, -100, I-Disease, -100, -100, -100].

BERT for Token Classification

The encoded input word-piece sequence is parsed into the *BertforTokenClassification* framework [5] which consists of the BERT Embedding Layer, BERT Encoder that consists of 12 Transformer encoder blocks (each encoder block with 12 attention heads, hidden size of 768, and 110 million parameters) and an additional Linear Classification Layer for fine-tuning.

In this research, we initialize the pre-trained model and weight from the food safety pre-trained BERT models, such as FS-BERT. The vectors of token representations are extracted from the input token sequences with the pre-trained BERT embeddings. The token representations are parsed into the 12 BERT Encoder layers to learn the contextual representations

between the tokens of the input sequence using the self-attention mechanism. Then, the contextual representation vectors are parsed into an additional Linear Classification Layer for the multi-class classification, to predict the NER labels of each token in the input sequence. Finally, the predicted sequence outputs are evaluated against the NER (true) labels using the standard evaluation metrics on precision, recall, and F1-score in an iterative manner to fine-tune the BERT model for the token classification task.

4.2.4 NER Fine-tuned BERT Models

Using the curated benchmark datasets given in Table 4.4, we perform the NER Task Specific Fine-tuning (TSF) on the vanilla BERT model [5] and the proposed domain adaptive pre-trained BERT models, as shown in Table 4.2. In this process, we train the pre-trained BERT models on the NER task to extract the "*Disease*", "*Food*", "*Location*", and "*Symptom*" named entities from the food safety datasets to augment the base dictionaries.

NER Model	Components	
	<i>DAP</i>	<i>TSF</i>
TSF-BERT	✗	✓
TSF-FS-BERT	✓	✓
TSF-FS-BERT _{News}	✓	✓
TSF-CRMS-BERT	✓	✓
TSF-CRMS-FS-BERT	✓	✓
TSF-CRMS-FS-BERT _{News}	✓	✓

Table 4.5: NER Fine-tuned BERT Models.

Table 4.5 shows the list of NER fine-tuned BERT models. In this research, we perform the same task specific fine-tuning process on the vanilla BERT model, without domain adaptive pre-training (DAP), to compare and evaluate the effectiveness of the DAP approach. We have developed six NER task specific fine-tuned models, namely TSF-BERT, TSF-FS-BERT, TSF-FS-BERT_{News}, TSF-CRMS-BERT, TSF-CRMS-FS-BERT, and TSF-CRMS-FS-BERT_{News}.

4.3 Performance Evaluation

In this section, we first discuss the experimental details which include the datasets, evaluation metrics and the implementation details. Then, we evaluate the performance of the NER approach based on the NER Benchmark Dataset, and the Food Safety Dataset.

4.3.1 Datasets

In this research, we evaluate the NER approach based on the NER Benchmark Dataset and the Food Safety Datasets, i.e., online food safely news articles and CRMS feedback texts.

NER Benchmark Dataset

Benchmark Dataset	Train	Validation	Test	#Total Sentences
NCBI (Disease) [65]	2,930	489	539	3,958
N2C2 [70]	3,128	782	2,605	6,515
FoodBase [79]	638	159	199	996
MIT_Restaurant [78]	1,121	280	277	1,678
CoNLL03 [16]	5,127	1,329	1,266	7,722
Total	12,944	3,039	4,886	20,869

Table 4.6: Statistics of the NER Benchmark Dataset.

We perform a train-validation-test dataset split of approximately 60:20:20 for the NER Benchmark Dataset for NER training. This mainly follows the dataset split ratio given by most benchmark datasets. Then, the data from the benchmark datasets are combined for NER fine-tuning. Table 4.6 shows the statistics of the NER Benchmark Dataset.

Food Safety Test Datasets

We evaluate the NER performance based on the "*Food Poisoning*" event related data from the food safety datasets on food safety news articles and CRMS feedback texts. Table 4.7

Food Safety Test Dataset	#Samples	#Sentences
Food Safety News Articles	40	826
CRMS Feedback	40	266

Table 4.7: Statistics of the food safety test dataset.

shows the statistics of the test datasets for food safety. A total of 40 news article samples with 826 sentences that are randomly selected from online food safety news articles with the "Food Poisoning" event type labels. Additionally, another 40 feedback case samples with 266 sentences that are randomly selected from the CRMS feedback cases by matching the feedback categories with the "food poisoning" keyword. The food safety test datasets are manually labeled for performance evaluation.

4.3.2 Evaluation Metrics

We use the standard evaluation metrics [99] that are commonly applied for NLP tasks to evaluate the performance of our NER models based on micro-averaged *precision*, *recall*, and *F1-score*. The evaluation metrics are defined as follows:

- Precision measures the proportion of correctly recognized named entities out of all the extracted named entities.

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \quad (4.1)$$

- Recall measures the proportion of correctly recognized named entities out of all relevant named entities.

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \quad (4.2)$$

- F1-score measures the overall performance of the NER model, accounting for both precision

and recall.

$$F1 = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4.3)$$

4.3.3 Implementation Details

We follow [58, 57] and apply the training hyperparameters for fine-tuning the pre-trained BERT models. We apply AdamW optimizer with 3e-5 learning rate and 0.01 weight decay. We train the BERT model [5] for 25 epochs with batch size 4. Random seeding is added to ensure the experiments are consistent and replicable. We evaluate the BERT model on the validation dataset after each epoch. We then save the model with the best F1-score.

4.3.4 Performance based on the NER Benchmark Dataset

We perform the Task Specific Fine-tuning (TSF) approach on the vanilla BERT model [5] and the proposed domain adaptive pre-trained BERT models using the NER Benchmark Dataset for low-resource NER modeling. The NER fine-tuned models shown in Table 4.5, namely TSF-BERT, TSF-FS-BERT, TSF-FS-BERT_{News}, TSF-CRMS-BERT, TSF-CRMS-FS-BERT, and TSF-CRMS-FS-BERT_{News}, are evaluated on the test set of the NER Benchmark Dataset.

NER Model	Precision	Recall	F1-score
TSF-BERT	83.02%	86.84%	84.89%
TSF-FS-BERT	75.24%	81.02%	78.03%
TSF-FS-BERT _{News}	79.27%	82.50%	80.86%
TSF-CRMS-BERT	82.34%	84.51%	83.41%
TSF-CRMS-FS-BERT	70.80%	81.67%	75.85%
TSF-CRMS-FS-BERT _{News}	78.32%	82.43%	80.33%

Table 4.8: Performance results based on the NER Benchmark Dataset.

The performance results based on the test set of the NER Benchmark Dataset are shown in Table 4.8. As can be seen, the TSF-BERT NER model, based on the vanilla pre-trained

BERT model [5], outperforms the NER models based on the proposed domain adaptive pre-trained BERT models on the NER Benchmark Dataset, achieving the performance of 84.89% in F1-score. Moreover, we compare the performance of the NER fine-tuned domain adaptive pre-trained models. We observe that the models that are pre-trained solely on the task-level corpora, i.e., TSF-FS-BERT_{News}, TSF-CRMS-BERT, and TSF-CRMS-FS-BERT_{News}, have achieved better performance results compared to the models that are pre-trained with both task-level and domain-level corpora, i.e., TSF-FS-BERT and TSF-CRMS-FS-BERT, which have achieved the performance of under 80% in F1-score. This may be due to the difference in word distribution in the larger pre-training corpora, which contains more specialized terms that are different from the distribution of terms found in the benchmark datasets.

4.3.5 Performance based on the Food Safety Test Datasets

We evaluate the performance of the proposed NER models separately based on the manually annotated food safety test datasets. We compare the performance results generated by the TSF-BERT NER model, based on the vanilla BERT model [5], and the NER fine-tuned models that are based on the proposed food safety domain adaptive pre-trained BERT models, as shown in Table 4.5. We aim to evaluate the effectiveness of the proposed domain adaptive pre-training (DAP) and task specific fine-tuning (TSF) approaches for food safety NER. Table 4.9 shows the overall performance results on the annotated Food Safety Test Datasets.

From the overall performance results based on the Food Safety Test Datasets, the TSF-BERT NER model, based on the vanilla pre-trained BERT model [5], has achieved better overall performance for both food safety news articles and CRMS feedback datasets. Moreover, we can also observe that the NER models that are pre-trained with both task-level and domain-level corpora, namely TSF-FS-BERT and TSF-CRMS-FS-BERT, exhibit significantly lower performance in comparison. This shows that further pre-training on larger

Dataset	NER Model	Precision	Recall	F1-score
Food Safety News Articles	TSF-BERT	48.70%	43.19%	45.78%
	TSF-FS-BERT	35.00%	35.18%	35.09%
	TSF-FS-BERT _{News}	42.15%	35.26%	38.40%
	TSF-CRMS-BERT	49.85%	40.30%	44.57%
	TSF-CRMS-FS-BERT	34.26%	35.59%	34.91%
	TSF-CRMS-FS-BERT _{News}	44.09%	39.14%	41.47%
CRMS Feedback	TSF-BERT	49.58%	42.89%	45.99%
	TSF-FS-BERT	35.26%	32.29%	33.71%
	TSF-FS-BERT _{News}	44.77%	33.01%	38.00%
	TSF-CRMS-BERT	46.26%	40.24%	43.04%
	TSF-CRMS-FS-BERT	36.28%	37.59%	36.92%
	TSF-CRMS-FS-BERT _{News}	41.30%	36.63%	38.83%

Table 4.9: Overall performance results based on the Food Safety Test Datasets for Named Entity Recognition (NER).

domain specific corpora (e.g., combining task-level and domain-level corpora for more content) introduces more specialized and unique terms that modify the contextual representations (embeddings) of the word-pieces in the existing pre-trained vocabulary of BERT [5], affecting the overall performance of the model.

Next, we further investigate the performance of the proposed NER models on individual NER categories. This is achieved by the evaluation metrics based on the recognition of individual category labels, namely "*Disease*", "*Food*", "*Location*", and "*Symptom*", from the Food Safety Test Datasets. In this research, we use the python framework for sequence labeling evaluation, called *segeval* [100].

Performance Results based on the "Disease" Category

Table 4.10 shows the performance results of the NER models on the "*Disease*" category. As can be seen, although the TSF-BERT NER model has achieved better overall performance, the TSF-CRMS-BERT NER model, based on the proposed domain adaptive pre-trained BERT model, has outperformed the TSF-BERT NER model for recognizing the specialized

Dataset	NER Model	Disease Entities		
		<i>Precision</i>	<i>Recall</i>	<i>F1-score</i>
Food Safety News Articles	TSF-BERT	42.58%	27.85%	33.67%
	TSF-FS-BERT	25.87%	31.22%	28.30%
	TSF-FS-BERT _{News}	35.94%	19.41%	25.21%
	TSF-CRMS-BERT	34.27%	41.35%	37.48%
	TSF-CRMS-FS-BERT	21.24%	27.43%	23.94%
	TSF-CRMS-FS-BERT _{News}	35.55%	38.40%	36.92%
CRMS Feedback	TSF-BERT	0%	0%	0%
	TSF-FS-BERT	0%	0%	0%
	TSF-FS-BERT _{News}	0%	0%	0%
	TSF-CRMS-BERT	5.56%	2.94%	03.85%
	TSF-CRMS-FS-BERT	0%	0%	0%
	TSF-CRMS-FS-BERT _{News}	0%	0%	0%

Table 4.10: Performance results of NER for the "*Disease*" category.

"*Disease*" named entities. This demonstrates the effectiveness of the DAP approach for recognizing terms in the specialized "*Disease*" category. We also observe poor performance for the CRMS feedback test dataset. This may be due to the small number of "*Disease*" named entities found in the annotated dataset. Only 34 named entity mentions are labeled as the "*Disease*" category out of 266 sentences in the CRMS feedback test dataset. Among them, 29 "*Disease*" named entities are the "*food poisoning*" term, which may be difficult to identify from the context, (e.g., the sentence "*Suspected food poisoning after consuming ...*" is annotated as ["*O*", "*B-Disease*", "*I-Disease*", "*O*", "*O*", ...] where "*food poisoning*" is difficult to recognize as a "*Disease*" in the context). The CRMS feedback test dataset contains a small number of specialized "*Disease*" terms, such as "*rotavirus*" and "*salmonella*". In addition, spelling errors could be found in the CRMS feedback dataset (e.g., "*samonella*" instead of "*salmonella*").

Performance Results based on the "Food" Category

Table 4.11 shows the performance results of the NER models on the "*Food*" category. As can be seen, the TSF-BERT NER model has performed significantly better than the other models

Dataset	NER Model	Food Entities		
		<i>Precision</i>	<i>Recall</i>	<i>F1-score</i>
Food Safety News Articles	TSF-BERT	64.59%	31.62%	42.45%
	TSF-FS-BERT	50.34%	17.10%	25.52%
	TSF-FS-BERT _{News}	61.64%	21.08%	31.41%
	TSF-CRMS-BERT	66.67%	28.10%	39.54%
	TSF-CRMS-FS-BERT	42.81%	28.57%	34.27%
	TSF-CRMS-FS-BERT _{News}	52.51%	26.93%	35.60%
CRMS Feedback	TSF-BERT	78.00%	50.32%	61.18%
	TSF-FS-BERT	64.18%	27.74%	38.74%
	TSF-FS-BERT _{News}	66.22%	31.61%	42.79%
	TSF-CRMS-BERT	71.43%	48.39%	57.69%
	TSF-CRMS-FS-BERT	54.46%	39.35%	45.69%
	TSF-CRMS-FS-BERT _{News}	58.88%	40.65%	48.09%

Table 4.11: Performance results of NER for the "Food" category.

for identifying the "Food" named entities from the food safety news article test dataset and the CRMS feedback test dataset.

Performance Results based on the "Location" Category

Dataset	NER Model	Location Entities		
		<i>Precision</i>	<i>Recall</i>	<i>F1-score</i>
Food Safety News Articles	TSF-BERT	71.12%	85.29%	77.56%
	TSF-FS-BERT	57.03%	68.95%	62.43%
	TSF-FS-BERT _{News}	58.06%	76.47%	66.01%
	TSF-CRMS-BERT	69.71%	77.45%	73.37%
	TSF-CRMS-FS-BERT	51.61%	67.97%	58.67%
	TSF-CRMS-FS-BERT _{News}	58.36%	69.61%	63.49%
CRMS Feedback	TSF-BERT	48.94%	20.72%	29.11%
	TSF-FS-BERT	28.81%	15.32%	20.00%
	TSF-FS-BERT _{News}	17.39%	10.81%	13.33%
	TSF-CRMS-BERT	38.96%	27.03%	31.91%
	TSF-CRMS-FS-BERT	19.51%	14.41%	16.58%
	TSF-CRMS-FS-BERT _{News}	17.65%	13.51%	15.31%

Table 4.12: Performance results of NER for the "Location" category.

Table 4.12 shows the performance results of the NER models on the individual "Location" category. Although the TSF-BERT NER model outperforms the other models in identifying

more generic *"Location"* named entities from the food safety news article test dataset (e.g., the names of countries, provinces, and states), the TSF-CRMS-BERT NER model performs significantly better in identifying Singapore-based *"Location"* named entities (e.g., *"Jurong West 505 Market"* and *"Clementi Ave"*) from the CRMS feedback test dataset, achieving the performance of 31.91% in F1-score, which is 2.8% better than the TSF-BERT NER model. This also demonstrates the effectiveness of the DAP approach for recognizing specialized *"Location"* terms.

Performance Results based on the *"Symptom"* Category

Dataset	NER Model	Symptom Entities		
		<i>Precision</i>	<i>Recall</i>	<i>F1-score</i>
Food Safety News Articles	TSF-BERT	17.78%	25.31%	20.89%
	TSF-FS-BERT	16.35%	28.22%	20.70%
	TSF-FS-BERT _{News}	16.96%	23.65%	19.76%
	TSF-CRMS-BERT	19.08%	13.69%	15.94%
	TSF-CRMS-FS-BERT	13.64%	14.94%	14.26%
	TSF-CRMS-FS-BERT _{News}	23.40%	22.82%	23.11%
CRMS Feedback	TSF-BERT	38.31%	66.96%	48.73%
	TSF-FS-BERT	29.72%	64.35%	40.66%
	TSF-FS-BERT _{News}	46.91%	66.09%	54.87%
	TSF-CRMS-BERT	37.89%	53.04%	44.20%
	TSF-CRMS-FS-BERT	34.50%	68.70%	45.93%
	TSF-CRMS-FS-BERT _{News}	43.27%	64.35%	51.75%

Table 4.13: Performance results of NER for the *"Symptom"* category.

Table 4.13 shows the performance results of the NER models on the *"Symptom"* category. As can be seen, the TSF-CRMS-FS-BERT_{News} NER model has achieved better performance in identifying *"Symptom"* named entities in the food safety news article test dataset, and the TSF-CRMS-BERT and TSF-CRMS-FS-BERT_{News} NER models have achieved better performance in the CRMS feedback test dataset. This demonstrates the effectiveness of the DAP approach for recognizing terms in the specialized *"Symptom"* category.

Performance Analysis

From the performance results, it has been shown that domain adaptive pre-training (DAP) can help recognize named entities from specialized categories, such as "*Disease*" and "*Symptom*", when pre-training on a specialized set of corpora to capture the context in the specialized datasets. For example, pre-training on the contents of CRMS feedback cases can improve the recognition of specialized "*Disease*", "*Location*", and "*Symptom*" named entities in the CRMS feedback test dataset. However, the pre-training approach to enhance the pre-trained BERT embeddings may affect the performance of the BERT NER model, which can be seen from the results on the food safety news article test dataset. Therefore, a trade-off can be observed when applying the DAP approach. In addition, we also observe that the use of larger pre-training corpora, i.e., the addition of domain-level corpora to task-level corpora, does not necessarily improve the pre-trained model's performance, which may be due to the difference in the word distribution of the corpora compared to the food safety dataset.

4.3.6 Performance on Based Dictionaries Augmentation

In this research, we also perform experiments on the NER module separately by extracting named entities from 200 "*food poisoning*" event related samples from each food safety dataset, based on each food safety category, including "*Disease*", "*Food*", "*Location*", and "*Symptom*", for base dictionaries augmentation. Table 4.14 shows the statistics of the food safety test datasets.

Food Safety Test Dataset	#Samples	#Sentence
Food Safety News Articles	200	4,277
CRMS Feedback	200	1,311

Table 4.14: Food safety test datasets for base dictionaries augmentation.

In our experiments, we use the TSF-FS-BERT and TSF-FS-BERT_{News} NER models

for extracting the named entities for the food safety news article test dataset, and use the TSF-CRMS-BERT, TSF-CRMS-FS-BERT, and TSF-CRMS-FS-BERT_{News} NER models for extracting named entities from the CRMS feedback test dataset. We extract the new "unseen" named entities by comparing the extracted entities from the base dictionaries. The extracted named entities are then manually curated and verified by the SFA scientist through the man-in-the-middle (MitM) process to form the augmented dictionaries. Table 4.15 shows the number of new entities extracted from the test datasets, in comparison to the base dictionaries, for each food safety category by the proposed NER models.

Dataset	NER Model	#New Named Entities			
		<i>Disease</i>	<i>Food</i>	<i>Location</i>	<i>Symptom</i>
External Resources	NEM _{Base}	138	5,866	453	328
Food Safety News Articles	TSF-BERT	+108	+278	+308	+47
	TSF-FS-BERT	+94	+165	+232	+48
	TSF-FS-BERT _{News}	+109	+201	+251	+46
	Total (Unique)	+159	+365	+343	+82
CRMS Feedback Text	TSF-BERT	+1	+122	+104	+58
	TSF-CRMS-BERT	+6	+138	+128	+56
	TSF-CRMS-FS-BERT	+4	+93	+76	+56
	TSF-CRMS-FS-BERT _{News}	+3	+114	+101	+58
	Total (Unique)	+7	+192	+205	+82

Table 4.15: Number of new entities extracted using the proposed NER models for base dictionaries augmentation.

From the results shown in Table 4.15 , we can observe that there are still a number of food safety related entities that are not captured by the base dictionaries. With only 200 food safety news articles and 200 CRMS feedback reports, we have extracted a total of 1,416 unique entities (948 entities from new articles and 486 entities from feedback texts, with 18 overlapping entities) from the NER models, showing an increase in the coverage of food safety dictionaries with the proposed NER models.

From this research, we can observe the usefulness of the proposed NER models for both food safety test datasets. From the food safety news articles test dataset, although the TSF-BERT NER model, based on the vanilla BERT model [5], has shown to extract more

"unseen" named entities from the "*Food*" and "*Location*" categories, the TSF-FS-BERT and TSF-FS-BERT_{News} NER models have shown better extraction for "unseen" named entities in the specialized "*Symptom*" and "*Disease*" categories, respectively. From the CRMS feedback test dataset, the TSF-CRMS-BERT NER model has shown to outperform all NER models for extracting "unseen" named entities from the "*Disease*", "*Food*", and "*Location*" categories, and both TSF-BERT and TSF-CRMS-FS-BERT_{News} NER models have shown to extract slightly more named entities from the "*Symptom*" category.

4.4 Summary

This chapter has discussed the proposed low-resource Named Entity Recognition (NER) approach to fine-tune the pre-trained BERT language model by leveraging multiple benchmark datasets from domains that are conceptually related to the food safety domain. The proposed NER approach performs transfer learning to recognize named entity for the "*Disease*", "*Food*", "*Location*", and "*Symptom*" categories in the food safety domain. In addition, we have also discussed the approach for Domain Adaptive Pre-training (DAP) to improve the domain adaptive capabilities of the pre-trained BERT model [5] to generalize better on the context of the target dataset, i.e., food safety news articles and customer feedback texts. It captures more specialized food safety related terms for the extraction of named entities to augment the base dictionaries. In total, we have extracted an additional 1,416 food safety named entities from 200 samples of each food safety test dataset. Among them, 948 named entities are extracted from food safety new articles, and 486 named entities are extracted from customer feedback texts, with 18 overlapping named entities. The proposed NER models can help reduce the laborious efforts and cost to manually extract named entities from the food safety datasets.

Chapter 5

Named Entity Matching (NEM)

The Named Entity Matching (NEM) module uses a heuristic dictionary-based approach to extract the named entities by matching strings (terms) from a given textual sequence with a pre-defined list of named entities in a dictionary [72]. It aims to achieve high precision in performance that are determined by the matching criteria for named entities and the quality and coverage of the named entities stored in the dictionaries. However, the heuristic dictionary-based approach commonly suffers from low recall as it is highly dependent on the completeness of the dictionaries [72, 101]. Therefore, in this research, we focus on curating a comprehensive list of dictionaries for the named entity extraction task, as discussed in the previous chapters.

Figure 5.1 shows the proposed Named Entity Matching (NEM) approach for the NEE-FSM framework for food safety monitoring. First, new contents of the input data from the SFA Food Safety Monitoring (SFA-FSM) system, i.e., online food safety news articles and CRMS customer feedback texts, are split into sentence-level inputs for named entity extraction. We then perform data cleaning to remove and replace uni-codes for standardization purposes. After that, we load the Augmented Dictionaries with the lists of named entities for each food safety category, namely "*Disease*", "*Food*", "*Location*", and "*Symp-*

5.1 Base and Augmented Dictionaries

In this research, we have curated the base dictionaries using the Heuristic Entity Extraction (HEE) module by extracting lists of commonly known food safety related named entities from various knowledge resources. Then, we employ the Named Entity Recognition (NER) module to extract new and "unseen" food safety named entities from additional food safety datasets to augment the base dictionaries.

Dictionaries	Resource	#Named Entities				
		<i>Disease</i>	<i>Food</i>	<i>Location</i>	<i>Symptom</i>	<i>Total</i>
Base Dictionaries	External Knowledge Resource	138	5,866	453	328	6,785
Augmented Dictionaries	200 Food safety News Articles	297	6,230	796	410	7,733
	200 CRMS Feedback	145	6,058	658	410	7,271
	Total Data	303	6,418	998	482	8,202

Table 5.1: Base and augmented dictionaries.

Table 5.1 shows the statistics of the named entities for each food safety category, namely "*Disease*", "*Food*", "*Location*", and "*Symptom*", of the base and augmented dictionaries. The augmented dictionaries consist of the named entities from the base dictionaries and the named entities extracted by the NER models from 200 samples of each food safety data, namely the food safety news articles and CRMS feedback texts.

5.2 n -gram Dictionaries Generation

In the NEM extraction process, we first sort and group the named entities from each food safety category dictionaries by the length of each named entity term with n -gram. Next, we split the dictionaries into smaller in-memory lists of named entities, called the n -gram dictionaries. Then, we order the n -gram dictionaries by the longest n -gram for matching. The n -gram dictionaries are used by the NEM to match the longest term first, in order to reduce the textual ambiguities and noisy extractions. This is due to the various overlappings of the food safety named entities in each dictionary. For example, if given an input sentence

"... suspected food poisoning when she ate chicken rice ...", and the "Food" entity dictionary consisting of the list of named entities, e.g., ["chicken rice", "chicken", "rice"], we aim to extract the longest entity "chicken rice" for the "Food" named entity instead of extracting all its permutations, such as "chicken" and "rice", as these are considered as noise that may affect the performance of downstream tasks.

Category	<i>n</i> -gram	#Named Entities
Disease	8-gram	1
	6-gram	2
	5-gram	2
	4-gram	11
	3-gram	66
	2-gram	116
	1-gram	105
Food	9-gram	1
	6-gram	12
	5-gram	42
	4-gram	199
	3-gram	987
	2-gram	3,614
	1-gram	1,563
Location	8-gram	1
	7-gram	4
	6-gram	7
	5-gram	16
	4-gram	63
	3-gram	178
	2-gram	265
	1-gram	464
Symptom	6-gram	2
	5-gram	1
	4-gram	14
	3-gram	19
	2-gram	139
	1-gram	307

Table 5.2: Statistics on *n*-gram terms for each food safety category.

Table 5.2 shows the statistics on *n*-gram terms for each food safety category from the augmented dictionaries. As can be seen, longer *n*-gram terms are rare among the list of named entities in comparison to uni-gram (1-gram) and bi-gram (2-gram) terms. Therefore,

the extraction can be more precise for matching the longer terms. For example, the "*Disease*" named entity with an 8-gram term, such as "*shiga toxin-producing e. coli (stec) infections*" or a 6-gram term, such as "*e. coli o157 : h7 infections*" are more specific compared to the bi-gram term "*e. coli*".

5.3 Entity Matching

To perform named entity extraction, we first split the input strings into sentence-level inputs. Then, for each food safety category, we perform entity extraction by using the sorted n -gram dictionaries as look-up tables to extract the entity mentions starting from the longest n -gram dictionary (in-memory list). we employ the Regular Expression (RegEx) to perform exact matching with the "*OR*" operator that is delimited by the "|" symbol, e.g., "*r\b(...|bao|burger|burgers|burrata|cakes|...)\b*" is an example of the RegEx pattern for exact matching of the "*Food*" entity such as "*bao*", "*burger*", or "*burrata*", etc. We exhaustively extract all entity mentions and positional information from the input sentences. Then, using the positional information, we compare the overlapping terms to identify the longest term to be extracted.

For example, if given a sentence "*... suspected food poisoning when she ate chicken rice ...*", where the "*Food*" entities "*chicken rice*", "*chicken*", and "*rice*" are exhaustively extracted by matching the n -gram dictionaries. The positions of the terms "*chicken*" (8,8) and "*rice*" (9,9), fall within the position of the term "*chicken rice*" (8,9) in the sentence. Therefore "*chicken rice*" is extracted as the "*Food*" named entity.

5.4 Performance Evaluation

In this section, we evaluate the proposed named entity extraction for food safety monitoring (NEE-FSM) framework on two newly annotated food safety test datasets on food safety articles and CRMS feedback texts.

5.4.1 Food Safety Test Dataset

Dataset	#Articles	#Sentences
Food Safety News Articles	40	876
CRMS Feedback	40	241

Table 5.3: Curated NEM base dictionaries.

We evaluate the performance of NEM based on the newly annotated set of "*Food Poisoning*" event related test data from food safety news articles and CRMS feedback text. Table 5.3 shows the food safety test datasets which consist of 40 food safety news articles with 876 sentences, and 40 CRMS feedback cases with 241 sentences that are randomly selected and annotated for testing purposes. In this research, we aim to evaluate the performance of NEM extraction using the base dictionaries, and the augmented dictionaries.

5.4.2 Experiments

We first perform the named entity matching (NEM) approach using the base dictionaries and augmented dictionaries. Then, we evaluate the named entity extraction task as a sequence labeling task, by converting the extracted named entities into a word-level labeled sequence using the positional information and the BIO-tagging scheme [98]. For example, the sentence "... suspected food poisoning when she ate chicken rice ...", where "*chicken rice*" (8,9) is extracted as a "*Food*" entity and "*Food Poisoning*" (3,4) is extracted as a "*Disease*" entity, which will be converted into the word-level BIO-tag sequence [..., "O", "B-Disease", "I-

Disease", "O", "O", "O", "B-Food", "I-Food", ...]. Finally, using the annotated (true) label sequence as reference, we compute the performance results using the standard evaluation metrics for the micro-averaged *precision*, *recall*, and *F1-score*.

5.4.3 Performance Results

Dataset	Dictionaries	Precision	Recall	F1-score	Δ F1-score
Food Safety News Articles	Base Dictionaries	56.19%	30.70%	39.71%	+36.62%
	Augmented Dictionaries	69.58%	84.54%	76.33%	
CRMS Feedback	Base Dictionaries	52.77%	27.71%	36.34%	+37.62%
	Augmented Dictionaries	73.89%	74.03%	73.96%	

Table 5.4: Overall performance results for Named Entity Extraction approach.

Table 5.4 shows the overall performance results of the Named Entity Extraction for Food Safety Monitoring (NEE-FSM) framework. The Δ F1-score refers to the difference in F1-scores achieved by the augmented dictionaries when compared with the base dictionaries. The results show a significant improvement in performance when using the augmented dictionaries compared to the base dictionaries on both target datasets. As can be seen, the performance results show the effectiveness of the increased coverage of named entity dictionaries using the NER dictionary augmentation. We can observe significant improvements of 50.08% in recall on the food safety test datasets, while maintaining high precision using the NEM approach. Overall, the proposed named entity extraction has achieved a promising performance of 75.15% in micro-averaged F1-score for the food safety news articles and CRMS feedback datasets.

Performance Results based on the "Disease" Category

Table 5.5 shows the performance results of named entity extraction for the "Disease" category using the base dictionaries and the augmented dictionaries. As can be seen, the performance results show that the "Disease" named entity extraction achieves the highest

Dataset	Dictionaries	Disease Entities			
		<i>Precision</i>	<i>Recall</i>	<i>F1-score</i>	Δ F1-score
Food Safety News Articles	Base Dictionaries	86.49%	12.12%	21.26%	+60.06%
	Augmented Dictionaries	78.72%	84.09%	81.32%	
CRMS Feedback	Base Dictionaries	0%	0%	0%	+98.70%
	Augmented Dictionaries	100%	97.44%	98.70%	

Table 5.5: Performance results for Named Entity Extraction for "*Disease*" category.

improvements compared to the other food safety categories, with an average of 79.38% improvement in F1-score. Therefore, the results show that the base dictionaries constructed using the SFA provided food-borne and zoonotic "*Disease*" names are insufficient to perform the extraction from the food safety test datasets.

To elaborate, we observe low recall performance and relatively high precision achieved by the base dictionaries on the food safety news articles dataset. This shows that the Base Dictionaries do not contain the "*Disease*" names to match entity mentions from the dataset, and the "*Disease*" names that have been matched have achieved an 86.49% in precision score. Moreover, with more variations of "*Disease*" named entities from the augmented dictionaries available for entity matching, lower precision can be observed. The 0% precision and recall achieved on the CRMS feedback dataset show that the base dictionaries do not contain any "*Disease*" names from the CRMS feedback dataset. However, the 100% precision achieved shows that the augmented dictionaries provide complete match of "*Disease*" names from the CRMS feedback dataset.

Performance Results based on the "Food" Category

Table 5.6 shows the performance results of named entity extraction for the "*Food*" category. As can be seen, the performance results show an average improvement of 30.11% in F1-score for the extraction of "*Food*" named entities using the augmented dictionaries. We can observe fairly similar performance has been achieved for both the food safety news articles

Dataset	Dictionaries	Food Entities			
		<i>Precision</i>	<i>Recall</i>	<i>F1-score</i>	Δ F1-score
Food Safety News Articles	Base Dictionaries	42.57%	41.65%	42.11%	+25.52%
	Augmented Dictionaries	59.70%	78.00%	67.63%	
CRMS Feedback	Base Dictionaries	38.67%	39.77%	39.22%	+34.70%
	Augmented Dictionaries	66.67%	82.95%	73.92%	

Table 5.6: Performance results for Named Entity Extraction for "*Food*" category.

and CRMS feedback datasets. This shows that the lists of "*Food*" names can be generically used between the datasets. Additionally, we have observed that the base dictionaries for "*Food*" named entities achieved better recall performance compared to the other food safety categories, with an average of 40.71%. This shows higher coverage of "*Food*" entities curated from the community databases and Wikipedia through the HEE module.

Performance Results based on the "Location" Category

Dataset	Dictionaries	Location Entities			Δ F1-score
		<i>Precision</i>	<i>Recall</i>	<i>F1-score</i>	
Food Safety News Articles	Base Dictionaries	89.63%	33.80%	49.09%	+36.04%
	Augmented Dictionaries	78.09%	93.58%	85.13%	
CRMS Feedback	Base Dictionaries	60.00%	1.89%	3.66%	+37.85%
	Augmented Dictionaries	60.55%	41.51%	49.25%	

Table 5.7: Performance results for Named Entity Extraction for "*Location*" category.

Table 5.7 shows the performance results of named entity extraction for the "*Location*" category. From the improvements in the performance results of the "*Location*" named entity extraction from the food safety test datasets, we can observe that the NER models are able to capture "*Location*" terms, such as regions, provinces, states, and places, in addition to the names of countries in the base dictionaries. This is especially true for the extraction of Singapore-based "*Location*" named entities, with a 37.85% improvement in F1-score. However, poorer performance results (49.25% in F1-score) can be observed for extracting "*Location*" names in the CRMS feedback dataset compared to extracting "*Location*" names from the food safety news articles. This shows that extracting Singapore-based "*Location*"

named entities is a challenging task.

Performance Results based on the "Symptom" Category

Dataset	Dictionaries	Symptom Entities			Δ F1-score
		Precision	Recall	F1-score	
Food Safety News Articles	Base Dictionaries	68.04%	24.18%	35.68%	+41.60%
	Augmented Dictionaries	70.61%	85.35%	77.28%	
CRMS Feedback	Base Dictionaries	82.35%	49.30%	61.67%	+28.43%
	Augmented Dictionaries	87.42%	92.96%	90.10%	

Table 5.8: Performance results for Named Entity Extraction for "*Symptom*" category.

Table 5.8 shows the performance results of named entity extraction for the "*Symptom*" category. As can be seen, the performance results show a significant 41.60% improvement in F1-score for the extraction of "*Symptom*" entities from the food safety news articles test dataset, compared to the lower 28.43% improvement in F1-score from the CRMS feedback test dataset. The high performance results achieved for the CRMS feedback dataset by using the base dictionaries (61.67% in F1-score) and augmented dictionaries (90.10% in F1-score) show that the dictionaries provide good coverage for "*Symptom*" named entities.

5.4.4 Error Analysis

In this research, we observe three different scenarios in which the performance of named entity extraction may be affected by the quality of the dictionaries. These examples are sampled from the food safety news dataset during the evaluation process.

First, we observe ambiguous terms from the dictionaries that are categorized as different food safety categories. For example, depending on the context, the "*gastrointestinal illnesses*" and "*vesicular stomatitis*" terms can be recognized as either "*Disease*" or "*Symptom*" entities, and "*turkey*" can be recognized as either a "*Food*" or "*Location*" entity. Next, we observe fragmented named entities from the dictionaries, where the named entities in

the dictionaries are fine-grained and general. For example, if given the sentence "*E. coli O103, one of the more rare Shiga-toxin producing E. coli ...*", the "*e. coli*" term is extracted instead of the full terms "*e. coli O103*" and "*shiga-toxin producing e. coli*". And lastly, we observe false positive extractions as the NEM does not consider the context of the input sentences. For example, if given the sentence "... the dairy is still not producing products for sale .", the term "*dairy*" is wrongly extracted as a "*Food*" entity. Similarly, in the sentence "*to determine if a sick person is part of an outbreak .*", the term "*sick*" is wrongly extracted as a "*Symptom*" entity. Therefore, we can observe that the coverage and quality of named entities in the dictionaries may affect the performance of the dictionary matching.

5.5 Summary

This chapter has discussed the proposed dictionary-based Named Entity Matching (NEM) approach for Named Entity Extraction for the Food Safety Monitoring (NEE-FSM) framework. Although its design is simple and straight-forward, the NEM module has demonstrated its effectiveness for named entity extraction. However, the performance of the dictionary-based approach is restricted by the coverage of the named entity dictionaries. In addition, we have also shown the effectiveness of dictionary augmentation by using the neural Named Entity Recognition (NER) approach. With a total of 1,417 named entities extracted by the NER module, on top of the 6,785 named entities from the base dictionaries (8,202 named entities in total), we have achieved an average of about 75.15% performance in F1-score on both food safety datasets. In summary, the proposed NEE-FSM framework has shown that the incorporation of neural NER dictionary augmentation is a feasible approach to enhance the coverage of food safety named entities, showing significant improvements of about 50.08% in recall and 37.12% in F1-score, while maintaining high precision for entity extraction using the dictionary-based NER approach.

Chapter 6

Conclusion and Further Work

6.1 Conclusion

In collaboration with the Singapore Food Agency (SFA), we have developed the predictive Food Safety Monitoring system, called SFA-FSM. The SFA-FSM system incorporates advanced analytics and artificial intelligence (AI) functionalities to facilitate various aspects of SFA operations. The SFA-FSM system comprises three main components: Online Food Safety News Monitoring, Food Safety Feedback Monitoring, and Named Entity Extraction for food safety monitoring.

This thesis presents the Named Entity Extraction for Food Safety Monitoring (NEE-FSM) framework, which supports multiple downstream functionalities of the proposed SFA-FSM system. These functionalities include the food safety attribute extraction for both online food safety news articles and customer feedback texts. The NEE-FSM framework comprises three modules: the Heuristic Entity Extraction (HEE) module, which constructs the base dictionaries with relatively high quality by extracting commonly known food safety named entities from various knowledge resources such as community databases, Wikipedia,

and WordNet. The Named Entity Recognition (NER) module, which extracts new "unseen" food safety named entities from the food safety datasets, namely the food safety news article dataset and CRMS feedback dataset, to augment and enhance the coverage of the base dictionaries. And the Named Entity Matching (NEM) module, which performs the main food safety named entity extraction task using a heuristic dictionary-based approach with the augmented dictionaries. Additionally, we introduce a man-in-the-middle (MitM) dictionary curation process that involves the assistance of SFA scientists to ensure the quality of the dictionaries.

The proposed NEE-FSM framework enhances the low recall issues commonly associated with dictionary-based approaches by leveraging NER extractions to augment the base dictionaries, while maintaining high precision in named entity extractions with the dictionary-based NEM extractions. In this research, we have extracted 1,417 food safety named entities from 200 samples of each food safety dataset using the constructed NER models. These extractions are used to augment the base dictionaries, which contain 6,785 commonly known food safety named entities, resulting in a total of 8,202 unique named entities for the "*Disease*", "*Food*", "*Location*", and "*Symptom*" food safety categories. The overall performance results for the NEE-FSM framework have shown promising performance, averaging about 75.15% in F1-score on the food safety datasets for named entity extraction. We have achieved a significant improvement of about 50.08% in recall when using the augmented dictionaries compared to the base dictionaries, demonstrating the effectiveness of the proposed NER dictionary augmentation approach.

6.2 Future Work

The proposed NEE-FSM framework has shown feasible approaches for leveraging various publicly available knowledge resources to alleviate low-resource issues and reduce the heavy

reliance on specialized domain knowledge for manual data processing. Therefore, in future work, we can leverage this research to expand the coverage to include more food safety named entity categories, such as "*Animal*", "*Plant*", "*Brand*", "*Organization*", "*Causative agent*", and "*Pathogen*" to cater to other food safety event types such as "*Food Recall*" and "*Zoonotic Disease*" for a comprehensive and holistic food safety analysis.

Inspired by related works that enhance dictionary-based named entity extraction architectures with neural components as a hybrid framework, we can consider future directions to improve the research reported in this thesis. One possible approach is to apply a context-aware component [48] as an added intelligence layer to the heuristic dictionary-based NEM extraction approach. This involves leveraging pre-trained language models to exploit their pre-trained embeddings and consider relationships and contextual information between the named entities from an input sequence, along with the named entity dictionaries. The goal is to enhance the quality of NEM extractions from the input sequences, including textual disambiguation, negation text handling, and minimizing post-processing efforts for downstream systems.

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